

Finite-Time and Stationary Mean-Field Analysis for an Underdamped Diffusion with Row-Normalized Alignment

Abstract

We study an underdamped Langevin system on $\mathbb{R}^d$ with strong confinement and a row-normalized viscous alignment force. The central analytic difficulty is that the interaction is normalized by a state-dependent denominator, while the velocity drift remains nonlinear and nonlocal at the mean-field level. We first prove the structural estimates that make the model tractable: the normalized alignment force is uniformly bounded, dissipative on relative kinetic disagreement, and stable under differentiation. We then establish finite- N and McKean–Vlasov well-posedness, quantitative finite-time mean-field approximation on compact time intervals, and stationary thermodynamic compactness of the finite-particle invariant laws. In a perturbative small-viscosity regime we prove uniqueness of the stationary mean-field law and a stationary mean-field logarithmic Sobolev inequality. The result is a comprehensive finite-time and stationary mean-field analysis for an underdamped diffusion with row-normalized alignment.

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1 Introduction

We study a confining underdamped diffusion with row-normalized velocity alignment. The model couples transport in position with Langevin friction and noise in velocity, together with a nonlocal alignment drift whose normalization depends on the current spatial configuration. The normalization is the main analytic feature of the problem: it improves stability by averaging local velocity disagreement, but it also introduces a state-dependent denominator that must be controlled uniformly in both the particle system and the McKean–Vlasov limit.

The model studied here is the following.

- The position variable evolves by transport: $\dot{x} = v$.
- The velocity variable is subject to a confining force $-\nabla U(x)$, linear friction $-\gamma v$, isotropic Brownian noise, and a row-normalized velocity alignment force.
- After each observation interval one may apply a smooth radial squashing map ψ to cap the observed speed without destroying the previously obtained probabilistic estimates.

Three pillars organize the analysis.

1. *Structural control of the normalized alignment force.* The row-normalized viscous alignment is uniformly bounded, dissipative on relative kinetic disagreement, and perturbative relative to the hypocoercive backbone.
2. *Finite-time mean-field approximation.* The finite- N and McKean–Vlasov dynamics are globally well posed, and the normalized interaction remains compatible with quantitative propagation of chaos on compact time intervals.
3. *Stationary thermodynamic limit and perturbative stationary regularity.* Finite-particle invariant laws satisfy Harris-type compactness estimates; their stationary limit is identified with the unique stationary McKean–Vlasov law in a perturbative small-viscosity regime; and the stationary mean-field law satisfies a stationary mean-field logarithmic Sobolev inequality in the perturbative small-viscosity regime.

The paper is organized accordingly. Section 2 introduces the model and records the observation-level squashing map. Section 3 develops the structural bounds for the normalized alignment force. Section 4 establishes the one-particle hypocoercive backbone, and Section 5 upgrades that structure to the tensorized reference setting used later in the particle and stationary arguments. Section 6 proves the quantitative finite-time mean-field limit by synchronous coupling and path-space entropy. Section 7 develops the stationary theory: finite-particle invariant laws are constructed by Harris theory, stationary subsequential limits are identified, uniqueness is proved in a perturbative small-viscosity regime, and the perturbative stationary mean-field LSI is obtained by transfer from the frozen linear generator. The analysis uses only tools from Langevin dynamics, hypocoercivity, Harris-type ergodic theory, and McKean–Vlasov limits [?, ?, ?, ?].

Main Contributions

1. We prove uniform structural estimates for the row-normalized alignment force: boundedness, dissipativity, derivative control, and frozen-position Gaussian coercivity.
2. We establish finite-time quantitative propagation of chaos, both in a strong synchronous-coupling form and in an information-theoretic path-space relative entropy form.
3. We prove stationary thermodynamic compactness and identify every stationary subsequential limit as a stationary McKean–Vlasov law.
4. In a perturbative small-viscosity regime we prove uniqueness of the stationary mean-field law and a stationary mean-field logarithmic Sobolev inequality.

Main Theorem Bundle

Under Assumptions 2.1 and 2.3, the paper proves the following theorem families.

1. The finite- N system and the McKean–Vlasov equation are globally well posed and propagate quadratic moments.
2. On every compact time interval, the empirical law converges quantitatively to the nonlinear law, both in strong synchronous coupling and in path-space relative entropy.
3. Finite-particle invariant laws exist for every N , and every stationary subsequential limit is a stationary McKean–Vlasov law.
4. In the perturbative small-viscosity regime, the stationary McKean–Vlasov law is unique and satisfies a perturbative stationary mean-field LSI.

Dependency Roadmap

The table below records the main logical dependencies of the theorem chain and isolates the only results that require the perturbative small-viscosity regime.

Result	Main Inputs	Perturbative small-viscosity regime?
Finite- N well-posedness and moments	Proposition 3.5, Proposition 3.6, structural force bounds	no
McKean–Vlasov well-posedness and stability	Lemma 3.4, Proposition 3.5, Corollary 3.7	no
Strong propagation of chaos	Lemma 6.1, Lemma 6.3, Corollary 3.7, Theorem 6.4	no
Stationary subsequential limit	Theorem 7.4, Theorem 7.5, Theorem 7.9, Theorem 7.15	no
Uniqueness of the stationary mean-field law	Proposition 7.10, Lemma 7.12, Lemma 7.13, Proposition 7.14	yes
Perturbative stationary mean-field LSI	Definition 7.17, Proposition 7.18, Theorem 7.19	yes

Perturbative Small-Viscosity Regime

Only the stationary uniqueness theorem and the perturbative stationary mean-field LSI require the perturbative small-viscosity regime. All finite-time results, as well as the existence of finite-particle invariant laws and the stationary subsequential compactness theory, are unconditional under the standing assumptions. The threshold ν_* is introduced where the alignment-map contraction is proved; it depends only on the structural parameters of the model and on the contraction constants appearing in Lemma 7.13 and Proposition 7.18.

Remark 1.1 (Scope of the assumptions). The standing assumptions are chosen to give a clean finite-time and stationary theory with explicit hypocoercive and Harris-type inputs. In particular, the uniform convexity of U is used to obtain both the backbone entropy gap and the stationary compactness estimates. Optimization of assumptions is not pursued here.

2 Model and Main Assumptions

2.1 Standing assumptions

Assumption 2.1 (Strongly confining potential). The potential $U : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies:

(i) $U \in C^2(\mathbb{R}^d)$ and there exist constants $0 < m_U \leq L_U < \infty$ such that

$$\|\nabla U(x) - \nabla U(y)\| \leq L_U \|x - y\| \quad \forall x, y \in \mathbb{R}^d.$$

(ii) The Hessian is uniformly positive in the sense

$$\langle \xi, \nabla^2 U(x) \xi \rangle \geq m_U \|\xi\|^2 \quad \forall x, \xi \in \mathbb{R}^d.$$

(iii) U is coercive in the Lyapunov sense:

$$\langle x, \nabla U(x) \rangle \geq \alpha_U \|x\|^2 - \beta_U \quad \forall x \in \mathbb{R}^d$$

for some $\alpha_U > 0$ and $\beta_U < \infty$.

Remark 2.2. Assumption 2.1 for items (i) and (ii) places the paper in the classical uniformly convex Langevin regime. In this regime the Gibbs measure (7) satisfies a logarithmic Sobolev inequality by the Bakry-Émery criterion, and the modified-entropy computation below yields an explicit hypocoercive decay rate for the kinetic backbone; see [?, ?].

Assumption 2.3 (Regularized Gaussian kernel). Fix $\ell_{\text{visc}} > 0$, $\kappa_0 > 0$, and define

$$K(x, y) := \kappa_0 + \exp\left(-\frac{\|x - y\|^2}{2\ell_{\text{visc}}^2}\right).$$

Then K is symmetric, strictly positive, bounded, and globally Lipschitz in each variable. Set

$$k_* := \inf_{x, y} K(x, y) = \kappa_0, \quad k^* := \sup_{x, y} K(x, y) = 1 + \kappa_0.$$

Definition 2.4 (Finite-particle viscous weights). For $N \geq 2$ and a configuration $X = (x_1, \dots, x_N) \in (\mathbb{R}^d)^N$, define

$$d_i(X) := \sum_{k \neq i} K(x_i, x_k), \quad \omega_{ij}(X) := \frac{K(x_i, x_j)}{d_i(X)} \quad (j \neq i).$$

Then $\omega_{ij} \geq 0$ and $\sum_{j \neq i} \omega_{ij} = 1$.

Definition 2.5 (Finite- N kinetic core). Fix $\gamma > 0$, $\sigma > 0$, and $\nu_{\text{visc}} \geq 0$. For $i = 1, \dots, N$, the finite-particle system is the Itô SDE

$$dX_i(t) = V_i(t) dt, \quad (1)$$

$$dV_i(t) = \left(-\nabla U(X_i(t)) - \gamma V_i(t) + F_i(X(t), V(t)) \right) dt + \sigma dW_i(t), \quad (2)$$

where the viscous force is

$$F_i(X, V) := \nu_{\text{visc}} \sum_{j \neq i} \omega_{ij}(X) (V_j - V_i). \quad (3)$$

The Brownian motions $W_1, \dots, W_N$ are independent standard d -dimensional Wiener processes.

Definition 2.6 (Mean-field viscous field). Let $f \in \mathcal{P}(\mathbb{R}^{2d})$ with spatial marginal

$$\rho_f(dx) := \int_{\mathbb{R}^d} f(dx, dq).$$

Define the normalized kernel mass

$$m_f(x) := \int_{\mathbb{R}^d} K(x, y) \rho_f(dy)$$

and the mean-field viscous field

$$F[f](x, v) := \frac{\nu_{\text{visc}}}{m_f(x)} \iint_{\mathbb{R}^{2d}} K(x, y) (q - v) f(dy, dq). \quad (4)$$

The mean-field kinetic equation is

$$\partial_t f_t + v \cdot \nabla_x f_t + \nabla_v \cdot \left[(-\nabla U(x) - \gamma v + F[f_t](x, v)) f_t \right] = \frac{\sigma^2}{2} \Delta_v f_t. \quad (5)$$

Remark 2.7. Because $m_f(x) \geq k_* > 0$ for every probability measure f , Definition 2.6 is globally well posed at the algebraic level: there is no denominator singularity in the mean-field alignment field.

Definition 2.8 (Velocity squashing). Fix $V_{\text{alg}} > 0$. The smooth velocity squashing map is

$$\psi(v) := V_{\text{alg}} \frac{v}{V_{\text{alg}} + \|v\|}, \quad \psi(0) := 0.$$

The associated observation map on phase space is

$$S(x, v) := (x, \psi(v)).$$

Definition 2.9 (BAOAB observation chain). Fix a time step $\Delta t > 0$. The algorithmic update used at observation times is the BAOAB Strang splitting of the kinetic core:

$$Z_{n+1}^{\text{pre}} = \Phi_B^{\Delta t/2} \circ \Phi_A^{\Delta t/2} \circ \Phi_O^{\Delta t} \circ \Phi_A^{\Delta t/2} \circ \Phi_B^{\Delta t/2}(Z_n),$$

where $Z_n = (X_n, V_n)$ and the substeps are the explicit maps

$$\Phi_B^h(X, V) := (X, V + h G(X, V)), \quad \Phi_A^h(X, V) := (X + h V, V).$$

Here $G(X, V) = (G_1(X, V), \dots, G_N(X, V))$ with

$$G_i(X, V) := -\nabla U(X_i) + F_i(X, V).$$

The Ornstein-Uhlenbeck substep is

$$\Phi_O^h(X, V) := \left(X, e^{-\gamma h} V + \sqrt{\frac{\sigma^2}{2\gamma}} (1 - e^{-2\gamma h}) \Xi \right),$$

where $\Xi \sim \mathcal{N}(0, I)$ is an independent standard Gaussian vector. The viscous force F_i is the row-normalized coupling from (3). The observed state is then postprocessed componentwise by

$$Z_{n+1} = S(Z_{n+1}^{\text{pre}}) = (X_{n+1}^{\text{pre}}, \psi(V_{n+1}^{\text{pre}})).$$

This is the BAOAB ordering used in the implementation.

Remark 2.10 (Status of the squashing map). The smooth squashing map is an observation-level operation, not part of the continuous-time diffusion that drives the finite-time and stationary mean-field analysis. It is recorded here because it preserves the probabilistic bounds needed when passing from the continuous kinetic dynamics to observed states. None of the main finite-time or stationary mean-field theorems depends on the squashing map.

Notation Guide

Notation	Meaning
K, k_*, k^*	regularized Gaussian kernel and its uniform lower/upper bounds
$d_i(X), \omega_{ij}(X)$	row-normalization denominator and normalized alignment weights
$F_i(X, V), F[f](x, v)$	finite-particle and mean-field viscous alignment drifts
$P_t^N, P_t^{(1),0}$	finite-particle semigroup and one-particle uncoupled backbone semigroup
Q_τ^N	observed BAOAB chain with postprocessed velocities
ρ_t^N, f_t	law of the finite-particle system and solution of the McKean–Vlasov equation
π_N, π_∞	finite-particle invariant law and stationary mean-field law
$\Gamma_{\text{hypo}}^N, \Gamma_{\text{hypo}}^\infty$	hypocoercive carré du champ at particle and mean-field levels
$P_{[0,T]}^N, Q_{[0,T]}^N$	path-space laws of the interacting and independent nonlinear systems

Section roadmap. Section 3 proves the structural estimates for the row-normalized alignment force. Section 4 establishes the one-particle hypocoercive backbone. Section 5 upgrades that backbone estimate to the tensorized reference structure. Section 6 proves the finite-time mean-field limit in strong coupling and path-space relative entropy. Section 7 then proves the stationary thermodynamic limit, uniqueness in the perturbative small-viscosity regime, and the perturbative stationary mean-field LSI.

3 Structural Bounds: Confinement, Viscosity, and Squashing

Lemma 3.1 (Smooth bounded cap). *The map ψ of Definition 2.8 is smooth on $\mathbb{R}^d \setminus \{0\}$, continuous on $\mathbb{R}^d$, satisfies $\|\psi(v)\| \leq V_{\text{alg}}$, and is 1-Lipschitz:*

$$\|\psi(v) - \psi(w)\| \leq \|v - w\| \quad \forall v, w \in \mathbb{R}^d.$$

Proof. Write $r = \|v\|$. For $v \neq 0$,

$$\psi(v) = \phi(r)v, \quad \phi(r) = \frac{V_{\text{alg}}}{V_{\text{alg}} + r}.$$

Hence

$$D\psi(v) = \phi(r)I + \phi'(r) \frac{v \otimes v}{r} = \frac{V_{\text{alg}}}{V_{\text{alg}} + r} I - \frac{V_{\text{alg}}}{(V_{\text{alg}} + r)^2} \frac{v \otimes v}{r}.$$

The radial eigenvalue is $V_{\text{alg}}^2/(V_{\text{alg}} + r)^2 \leq 1$ and each tangential eigenvalue is $V_{\text{alg}}/(V_{\text{alg}} + r) \leq 1$. Therefore

$$\|D\psi(v)\|_{\text{op}} \leq 1 \quad (v \neq 0).$$

Any $u \in \mathbb{R}^d$ decomposes as $u = \alpha v/r + u_{\perp}$ with $u_{\perp} \perp v$, so

$$\|D\psi(v)u\| \leq \frac{V_{\text{alg}}^2}{(V_{\text{alg}} + r)^2} |\alpha| + \frac{V_{\text{alg}}}{V_{\text{alg}} + r} \|u_{\perp}\| \leq \|u\|.$$

Hence ψ is 1-Lipschitz on $\mathbb{R}^d \setminus \{0\}$ by the mean-value theorem on line segments. At the origin,

$$\lim_{v \rightarrow 0} \|D\psi(v)\|_{\text{op}} = 1,$$

so the same Lipschitz bound extends continuously to $v = 0$. The speed cap is obvious from

$$\|\psi(v)\| = \frac{V_{\text{alg}}\|v\|}{V_{\text{alg}} + \|v\|} \leq V_{\text{alg}}.$$

□

Lemma 3.2 (Uniform bound on the finite-particle viscous force). *If $\max_{1 \leq k \leq N} \|V_k\| \leq V_*$, then for every i*

$$\|F_i(X, V)\| \leq 2\nu_{\text{visc}} V_*.$$

The bound is independent of N .

Proof. By Definition 2.4,

$$F_i(X, V) = \nu_{\text{visc}} \left(\sum_{j \neq i} \omega_{ij}(X) V_j - V_i \right).$$

Taking norms and using $\sum_{j \neq i} \omega_{ij} = 1$ gives

$$\|F_i(X, V)\| \leq \nu_{\text{visc}} \left(\sum_{j \neq i} \omega_{ij}(X) \|V_j\| + \|V_i\| \right) \leq 2\nu_{\text{visc}} V_*.$$

□

Lemma 3.3 (Dissipativity of the viscous coupling). *Fix the positions X and consider only the viscous ODE*

$$\dot{V}_i = \nu_{\text{visc}} \sum_{j \neq i} \omega_{ij}(X) (V_j - V_i).$$

Define

$$D(X) := \sum_{i=1}^N d_i(X), \quad \bar{V}_d := \frac{1}{D(X)} \sum_{i=1}^N d_i(X) V_i,$$

and the degree-weighted velocity variance

$$\mathcal{V}_d(V) := \frac{1}{D(X)} \sum_{i=1}^N d_i(X) \|V_i - \bar{V}_d\|^2.$$

Then

$$\frac{d}{dt} \mathcal{V}_d(V(t)) = -\frac{2\nu_{\text{visc}}}{D(X)} \sum_{1 \leq i < j \leq N} K(x_i, x_j) \|V_i - V_j\|^2 \leq 0. \quad (6)$$

Proof. Since X is frozen, the degrees $d_i(X)$ are constant in time. Set $\delta_i := V_i - \bar{V}_d$. By symmetry of K ,

$$\frac{d}{dt} \bar{V}_d = \frac{\nu_{\text{visc}}}{D(X)} \sum_{i=1}^N \sum_{j \neq i} K(x_i, x_j) (V_j - V_i) = 0,$$

so $\dot{\delta}_i = \dot{V}_i$. Therefore

$$\begin{aligned} \frac{d}{dt} \mathcal{V}_d(V(t)) &= \frac{2}{D(X)} \sum_{i=1}^N d_i(X) \langle \delta_i, \dot{V}_i \rangle \\ &= \frac{2\nu_{\text{visc}}}{D(X)} \sum_{i=1}^N \sum_{j \neq i} K(x_i, x_j) \langle \delta_i, \delta_j - \delta_i \rangle \\ &= -\frac{\nu_{\text{visc}}}{D(X)} \sum_{i \neq j} K(x_i, x_j) \|\delta_i - \delta_j\|^2 \\ &= -\frac{2\nu_{\text{visc}}}{D(X)} \sum_{1 \leq i < j \leq N} K(x_i, x_j) \|V_i - V_j\|^2, \end{aligned}$$

which is non-positive. $\square$

Lemma 3.4 (Lipschitz control of the mean-field alignment field). *Fix $M_2 < \infty$. Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^{2d})$ satisfy*

$$\int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) \mu(dx, dv) \leq M_2, \quad \int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) \nu(dx, dv) \leq M_2.$$

Then there exists a constant $L_{\text{mf}} = L_{\text{mf}}(M_2, \nu_{\text{visc}}, k_, k^*, \ell_{\text{visc}}) < \infty$ such that*

$$\|F[\mu](x, v) - F[\nu](x', v')\| \leq L_{\text{mf}} (\|x - x'\| + \|v - v'\| + W_1(\mu, \nu)),$$

where W_1 denotes the 1-Wasserstein distance on $\mathbb{R}^{2d}$.

Proof. Write

$$A[\mu](x) := \frac{1}{m_\mu(x)} \iint K(x, y) q \mu(dy, dq), \quad F[\mu](x, v) = \nu_{\text{visc}} (A[\mu](x) - v).$$

To estimate the denominator, use the coupling characterization of W_1 and the global Lipschitz bound on K :

$$|m_\mu(x) - m_\nu(x')| \leq \iint |K(x, y) - K(x', y')| \pi(dy, dq, dy', dq') \leq C_K (\|x - x'\| + W_1(\mu, \nu)),$$

for any coupling $\pi \in \Pi(\mu, \nu)$ and some constant C_K depending only on L_K . Since $m_\mu, m_\nu \geq k_*$, the reciprocal map satisfies

$$|m_\mu(x)^{-1} - m_\nu(x')^{-1}| \leq k_*^{-2} |m_\mu(x) - m_\nu(x')|.$$

For the numerator, fix a coupling $\pi \in \Pi(\mu, \nu)$ and write

$$J_\mu(x) := \iint K(x, y) q \mu(dy, dq).$$

Then

$$\|J_\mu(x) - J_\nu(x')\| \leq \iint \|K(x, y)q - K(x', y')q'\| d\pi.$$

By Cauchy–Schwarz, the second-moment hypothesis implies a uniform first-moment bound in the velocity variable on the class under consideration. Therefore the integrand is controlled by a constant times $\|x - x'\| + \|y - y'\| + \|q - q'\|$, so

$$\|J_\mu(x) - J_\nu(x')\| \leq C_J(\|x - x'\| + W_1(\mu, \nu)),$$

with C_J depending only on the kernel bounds and M_2 . Combining the numerator and denominator estimates, and using

$$A[\mu](x) - A[\nu](x') = \frac{J_\mu(x) - J_\nu(x')}{m_\mu(x)} + J_\nu(x')(m_\mu(x)^{-1} - m_\nu(x')^{-1}),$$

gives

$$\|A[\mu](x) - A[\nu](x')\| \leq C(\|x - x'\| + W_1(\mu, \nu)).$$

Finally,

$$F[\mu](x, v) - F[\nu](x', v') = \nu_{\text{visc}}(A[\mu](x) - A[\nu](x')) - \nu_{\text{visc}}(v - v'),$$

so

$$\|F[\mu](x, v) - F[\nu](x', v')\| \leq L_{\text{mf}}(\|x - x'\| + \|v - v'\| + W_1(\mu, \nu)). \quad \square$$

Proposition 3.5 (Finite- and infinite-dimensional well posedness). *Under Assumptions 2.1 and 2.3,*

- (i) *for every $N \geq 2$ the SDE (1)–(2) has a unique global strong solution;*
- (ii) *for every initial law $f_0 \in \mathcal{P}(\mathbb{R}^{2d})$ with finite second moment, the McKean-Vlasov equation (5) has a unique global solution $t \mapsto f_t$ in $C(\mathbb{R}_+, \mathcal{P}_2(\mathbb{R}^{2d}))$.*

Proof. Fix N and write the drift as

$$b_N(X, V) = (V, -\nabla U(X) - \gamma V + F(X, V)).$$

Because the kernel K is smooth and strictly positive, every denominator

$$d_i(X) = \sum_{k \neq i} K(x_i, x_k)$$

is bounded below by $(N - 1)k_*$, so each normalized weight $\omega_{ij}(X)$ is smooth in X . Hence b_N is locally Lipschitz on $(\mathbb{R}^{2d})^N$, and standard SDE theory gives a unique strong solution up to the explosion time. To rule out explosion, define

$$\mathcal{E}_N(X, V) := \frac{1}{N} \sum_{i=1}^N (\|X_i\|^2 + \|V_i\|^2).$$

Applying Itô's formula gives

$$d\mathcal{E}_N = \frac{2}{N} \sum_{i=1}^N \langle X_i, V_i \rangle dt + \frac{2}{N} \sum_{i=1}^N \langle V_i, -\nabla U(X_i) - \gamma V_i + F_i(X, V) \rangle dt$$

$$+ d\sigma^2 dt + \frac{2\sigma}{N} \sum_{i=1}^N \langle V_i, dW_i \rangle.$$

Use Young's inequality,

$$2\langle X_i, V_i \rangle \leq \|X_i\|^2 + \|V_i\|^2, \quad -2\langle V_i, \nabla U(X_i) \rangle \leq \frac{1}{m_U} \|V_i\|^2 + m_U \|\nabla U(X_i)\|^2,$$

and the linear-growth bound

$$\|\nabla U(x)\|^2 \leq 2L_U^2 \|x\|^2 + 2\|\nabla U(0)\|^2.$$

For the viscous term, Definition 2.4 and $\omega_{ij} \leq k^*/((N-1)k_*)$ imply

$$\begin{aligned} \sum_{i=1}^N \langle V_i, F_i(X, V) \rangle &\leq \nu_{\text{visc}} \sum_{i=1}^N |V_i| \left(\sum_{j \neq i} \omega_{ij}(X) |V_j| + |V_i| \right) \\ &\leq \nu_{\text{visc}} \sum_{i=1}^N |V_i|^2 + \frac{\nu_{\text{visc}}}{2} \sum_{i=1}^N \sum_{j \neq i} \omega_{ij}(X) (|V_i|^2 + |V_j|^2) \\ &\leq \nu_{\text{visc}} \left(\frac{3}{2} + \frac{k^*}{2k_*} \right) \sum_{i=1}^N |V_i|^2. \end{aligned}$$

Consequently

$$\frac{d}{dt} \mathbb{E} \mathcal{E}_N(t) \leq C_0 + C_1 \mathbb{E} \mathcal{E}_N(t)$$

with constants depending only on $(m_U, L_U, \gamma, \sigma, \nu_{\text{visc}}, k_*, k^*)$ and not on N . Grönwall's lemma then gives $\sup_{t \leq T} \mathbb{E} \mathcal{E}_N(t) < \infty$ on every finite interval, which excludes explosion. Hence the strong solution is global.

For the McKean-Vlasov equation, Lemma 3.4 gives the local Lipschitz dependence of the drift on (x, v, μ) on moment-bounded classes, while the force has at most linear growth in (x, v) under a finite second moment assumption on μ . A Picard iteration on the nonlinear SDE

$$d\bar{X}_t = \bar{V}_t dt, \quad d\bar{V}_t = (-\nabla U(\bar{X}_t) - \gamma \bar{V}_t + F[f_t](\bar{X}_t, \bar{V}_t)) dt + \sigma dW_t$$

combined with the same Lyapunov estimate as above yields a unique global strong solution with law $f_t \in C(\mathbb{R}_+, \mathcal{P}_2(\mathbb{R}^{2d}))$. $\square$

Proposition 3.6 (Quadratic moment bounds). *Let $(X_i, V_i)_{i=1}^N$ solve (1)–(2) and let f_t solve (5). Then for every finite horizon $T > 0$ there exists $C_T < \infty$, depending only on the model parameters and the initial second moments, such that*

$$\sup_{t \in [0, T]} \frac{1}{N} \sum_{i=1}^N \mathbb{E} [\|X_i(t)\|^2 + \|V_i(t)\|^2] \leq C_T$$

for every $N \geq 2$, and

$$\sup_{t \in [0, T]} \int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) f_t(dx, dv) \leq C_T.$$

Proof. For the finite-particle system, the estimate from Proposition 3.5 already gives a closed differential inequality for the averaged quadratic energy, so the uniform bound follows by taking expectations and applying Grönwall's lemma.

For the mean-field equation, apply Itô's formula to $\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2$ along the nonlinear SDE. Since

$$\|F[f_t](x, v)\| \leq \frac{\nu_{\text{visc}} k^*}{k_*} \left(\int \|q\| f_t(dy, dq) + \|v\| \right),$$

Young's inequality gives

$$\frac{d}{dt} \mathbb{E}[\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2] \leq C_0 + C_1 \mathbb{E}[\|\bar{X}_t\|^2 + \|\bar{V}_t\|^2].$$

Applying Grönwall's lemma yields the claimed bound. $\square$

Corollary 3.7 (Mean-field stability in W_2). *Let f_t and g_t be two solutions of (5) on $[0, T]$, with initial laws $f_0, g_0 \in \mathcal{P}_2(\mathbb{R}^{2d})$. Then there exists $C_T < \infty$, depending only on the model parameters and on the initial second moments, such that*

$$W_2(f_t, g_t)^2 \leq e^{C_T t} W_2(f_0, g_0)^2 \quad \forall t \in [0, T].$$

Proof. Let $(\bar{X}_t, \bar{V}_t)$ and $(\bar{Y}_t, \bar{W}_t)$ solve the two nonlinear SDEs driven by the same Brownian motion, with an optimal W_2 -coupling at time 0. Set

$$\Delta X_t := \bar{X}_t - \bar{Y}_t, \quad \Delta V_t := \bar{V}_t - \bar{W}_t.$$

Applying Itô's formula to $\|\Delta X_t\|^2 + \|\Delta V_t\|^2$ gives

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\|\Delta X_t\|^2 + \|\Delta V_t\|^2] &= 2\mathbb{E}\langle \Delta X_t, \Delta V_t \rangle - 2\gamma \mathbb{E}\|\Delta V_t\|^2 \\ &\quad - 2\mathbb{E}\langle \Delta V_t, \nabla U(\bar{X}_t) - \nabla U(\bar{Y}_t) \rangle \\ &\quad + 2\mathbb{E}\langle \Delta V_t, F[f_t](\bar{X}_t, \bar{V}_t) - F[g_t](\bar{Y}_t, \bar{W}_t) \rangle. \end{aligned}$$

Young's inequality controls the transport and confinement terms, while Lemma 3.4 and Proposition 3.6 yield

$$\|F[f_t](\bar{X}_t, \bar{V}_t) - F[g_t](\bar{Y}_t, \bar{W}_t)\| \leq L_T (\|\Delta X_t\| + \|\Delta V_t\| + W_2(f_t, g_t)),$$

for a constant $L_T < \infty$ depending only on T , the model parameters, and the propagated second moments. Since the chosen coupling gives

$$W_2(f_t, g_t)^2 \leq \mathbb{E}[\|\Delta X_t\|^2 + \|\Delta V_t\|^2],$$

another Young inequality yields

$$\frac{d}{dt} \mathbb{E}[\|\Delta X_t\|^2 + \|\Delta V_t\|^2] \leq C_T \mathbb{E}[\|\Delta X_t\|^2 + \|\Delta V_t\|^2].$$

Grönwall's lemma gives the claim. $\square$

4 The Hypocoercive Backbone

4.1 Weighted generator calculus for the uncoupled kinetic backbone

For the detailed hypocoercive calculations it is convenient to work first with the uncoupled kinetic backbone. Set

$$T := \frac{\sigma^2}{2\gamma}$$

and define the Gibbs density

$$\rho_*(x, v) := Z_*^{-1} \exp\left(-\frac{U(x) + \frac{1}{2}\|v\|^2}{T}\right). \quad (7)$$

Under the smooth confining assumptions on U , this is the invariant density of the one-particle underdamped Langevin diffusion.

Definition 4.1 (Relative density and hypocoercive functionals). Let f_t solve the uncoupled one-particle Fokker-Planck equation

$$\partial_t f_t + v \cdot \nabla_x f_t + \nabla_v \cdot ((\nabla U(x) + \gamma v) f_t) = \frac{\sigma^2}{2} \Delta_v f_t,$$

and write

$$h_t(x, v) := \frac{f_t(x, v)}{\rho_*(x, v)}.$$

For smooth positive h define

$$\mathcal{A}[h] := \int h \log h \, d\rho_*, \quad (8)$$

$$\mathcal{B}[h] := T \int \frac{|\nabla_v h|^2}{h} \, d\rho_* = T \int h |\nabla_v \log h|^2 \, d\rho_*, \quad (9)$$

$$\mathcal{C}[h] := T \int \frac{\nabla_x h \cdot \nabla_v h}{h} \, d\rho_* = T \int h \langle \nabla_x \log h, \nabla_v \log h \rangle \, d\rho_*, \quad (10)$$

$$\mathcal{D}[h] := T \int \frac{|\nabla_x h|^2}{h} \, d\rho_* = T \int h |\nabla_x \log h|^2 \, d\rho_*. \quad (11)$$

Proposition 4.2 (Exact generator decomposition). *Let L_0 denote the generator acting on h through*

$$\partial_t h = L_0 h = \rho_*^{-1} \partial_t (h \rho_*).$$

Then

$$L_0 = L_{\text{sym}} + L_{\text{anti}},$$

where

$$L_{\text{sym}} h = \gamma (T \Delta_v h - v \cdot \nabla_v h), \quad (12)$$

$$L_{\text{anti}} h = -v \cdot \nabla_x h + \nabla U(x) \cdot \nabla_v h. \quad (13)$$

Moreover:

- (i) L_{sym} is symmetric and non-positive in $L^2(\rho_*)$;

(ii) L_{anti} is anti-symmetric in $L^2(\rho_*)$.

Proof. By definition,

$$L_0 h = \frac{1}{\rho_*} \left[-v \cdot \nabla_x (h \rho_*) + \nabla_v \cdot ((\nabla U + \gamma v) h \rho_*) + \frac{\sigma^2}{2} \Delta_v (h \rho_*) \right].$$

We first expand the diffusion-friction part. Using

$$\nabla_v \rho_* = -\frac{v}{T} \rho_*, \quad \frac{\sigma^2}{2} = \gamma T,$$

we obtain

$$\begin{aligned} \nabla_v \cdot ((\nabla U + \gamma v) h \rho_*) + \gamma T \Delta_v (h \rho_*) &= \nabla U \cdot \nabla_v (h \rho_*) + \gamma \nabla_v \cdot (v h \rho_*) + \gamma T \Delta_v (h \rho_*) \\ &= \nabla U \cdot \nabla_v (h \rho_*) + \gamma \rho_* v \cdot \nabla_v h + \gamma d h \rho_* + \gamma v \cdot \nabla_v \rho_* h \\ &\quad + \gamma T \Delta_v (h \rho_*). \end{aligned}$$

The terms involving $v h \rho_*$ cancel after expanding $\Delta_v (h \rho_*)$ and using $\nabla_v \rho_* = -T^{-1} v \rho_*$. What remains is

$$\nabla U \cdot \nabla_v (h \rho_*) + \gamma T \nabla_v \cdot (\rho_* \nabla_v h).$$

Dividing by ρ_* gives

$$\nabla U \cdot \nabla_v h + \gamma (T \Delta_v h - v \cdot \nabla_v h).$$

The transport part expands as

$$-\frac{1}{\rho_*} v \cdot \nabla_x (h \rho_*) = -v \cdot \nabla_x h - v \cdot \nabla_x (\log \rho_*) h.$$

Since $\nabla_x \log \rho_* = -T^{-1} \nabla U$, the last term is $T^{-1} (v \cdot \nabla U) h$, which cancels exactly with the corresponding term created when expanding $\nabla U \cdot \nabla_v (h \rho_*) / \rho_*$. This proves (12)–(13).

For symmetry, let g, h be smooth compactly supported functions. Then

$$\int g L_{\text{sym}} h d\rho_* = \gamma T \int g \rho_*^{-1} \nabla_v \cdot (\rho_* \nabla_v h) d\rho_* = -\gamma T \int \nabla_v g \cdot \nabla_v h d\rho_*,$$

which is symmetric in (g, h) and non-positive on the diagonal.

For anti-symmetry, write

$$b(x, v) := (v, -\nabla U(x)).$$

Then

$$L_{\text{anti}} h = -b \cdot \nabla_{x,v} h.$$

It is enough to prove that $\nabla_{x,v} \cdot (b \rho_*) = 0$. Indeed,

$$\nabla_{x,v} \cdot (b \rho_*) = v \cdot \nabla_x \rho_* - \nabla U \cdot \nabla_v \rho_* = -\frac{1}{T} (v \cdot \nabla U) \rho_* + \frac{1}{T} (\nabla U \cdot v) \rho_* = 0.$$

Therefore

$$\int g L_{\text{anti}} h d\rho_* = - \int g b \cdot \nabla h \rho_* = \int h b \cdot \nabla g \rho_* = - \int h L_{\text{anti}} g d\rho_*. \quad \square$$

Proposition 4.3 (Commutators). *For smooth h ,*

$$[\nabla_v, L_0] h = -\gamma \nabla_v h - \nabla_x h, \quad (14)$$

$$[\nabla_x, L_0] h = (\nabla^2 U) \nabla_v h. \quad (15)$$

Proof. The symmetric part depends only on v , so $[\nabla_x, L_{\text{sym}}] = 0$ and

$$[\nabla_v, L_{\text{sym}}]h = \gamma(T\nabla_v \Delta_v h - \nabla_v(v \cdot \nabla_v h) + v \cdot \nabla_v \nabla_v h) = -\gamma \nabla_v h.$$

For the anti-symmetric part,

$$[\nabla_v, -v \cdot \nabla_x]h = -\nabla_x h, \quad [\nabla_v, \nabla U \cdot \nabla_v]h = 0,$$

which yields (14). Likewise,

$$[\nabla_x, -v \cdot \nabla_x]h = 0, \quad [\nabla_x, \nabla U \cdot \nabla_v]h = (\nabla^2 U) \nabla_v h,$$

which is (15). □

Proposition 4.4 (Exact differential identities). *Let h_t solve $\partial_t h_t = L_0 h_t$. Set*

$$M := \sup_{x \in \mathbb{R}^d} \|\nabla^2 U(x)\|_{\text{op}} \leq L_U.$$

Then:

$$\frac{d}{dt} \mathcal{A}[h_t] = -\gamma \mathcal{B}[h_t], \tag{16}$$

$$\frac{d}{dt} \mathcal{B}[h_t] \leq -2\gamma \mathcal{B}[h_t] - 2\mathcal{C}[h_t], \tag{17}$$

$$\frac{d}{dt} \mathcal{C}[h_t] \leq -\mathcal{D}[h_t] - \gamma \mathcal{C}[h_t] + M \sqrt{\mathcal{B}[h_t] \mathcal{D}[h_t]}, \tag{18}$$

$$\frac{d}{dt} \mathcal{D}[h_t] \leq 2M \sqrt{\mathcal{B}[h_t] \mathcal{D}[h_t]}. \tag{19}$$

Proof. Set $\ell_t := \log h_t$.

For (16), the anti-symmetric part drops out by Proposition 4.2, item *ii*, so

$$\frac{d}{dt} \mathcal{A}[h_t] = \int (1 + \log h_t) L_{\text{sym}} h_t d\rho_* = -\gamma T \int \frac{|\nabla_v h_t|^2}{h_t} d\rho_* = -\gamma \mathcal{B}[h_t].$$

We next differentiate

$$\mathcal{B}[h_t] = T \int h_t |\nabla_v \ell_t|^2 d\rho_*.$$

Since $\partial_t \ell_t = h_t^{-1} L_0 h_t$, the chain rule gives

$$\frac{d}{dt} \mathcal{B}[h_t] = \underbrace{T \int L_0 h_t |\nabla_v \ell_t|^2 d\rho_*}_{=: I_1} + \underbrace{2T \int h_t \nabla_v \ell_t \cdot \nabla_v (\partial_t \ell_t) d\rho_*}_{=: I_2}.$$

Because

$$\nabla_v (\partial_t \ell_t) = \nabla_v \left(\frac{L_0 h_t}{h_t} \right) = \frac{\nabla_v L_0 h_t}{h_t} - \frac{L_0 h_t}{h_t} \nabla_v \ell_t,$$

we obtain

$$I_2 = 2T \int \nabla_v \ell_t \cdot \nabla_v L_0 h_t d\rho_* - 2T \int L_0 h_t |\nabla_v \ell_t|^2 d\rho_*.$$

Therefore

$$\frac{d}{dt} \mathcal{B}[h_t] = 2T \int \nabla_v \ell_t \cdot \nabla_v L_0 h_t d\rho_* - T \int L_0 h_t |\nabla_v \ell_t|^2 d\rho_*.$$

Split $L_0 = L_{\text{sym}} + L_{\text{anti}}$ and treat both pieces separately.

For the symmetric piece, use

$$L_{\text{sym}} = \gamma T \nabla_v^* \nabla_v,$$

where ∇_v^* is the $L^2(\rho_*)$ adjoint of ∇_v . A direct Bakry-Émery computation yields

$$-2\gamma T^2 \int h_t \|\nabla_v^2 \log h_t\|_{\text{HS}}^2 d\rho_* \leq 0,$$

and the first-order contribution is exactly $-2\gamma \mathcal{B}[h_t]$.

For the anti-symmetric piece, integrate by parts using Proposition 4.2, item *ii*:

$$\begin{aligned} 2T \int \nabla_v \ell_t \cdot \nabla_v L_{\text{anti}} h_t d\rho_* - T \int L_{\text{anti}} h_t |\nabla_v \ell_t|^2 d\rho_* \\ = 2T \int h_t \nabla_v \ell_t \cdot [\nabla_v, L_{\text{anti}}] \ell_t d\rho_*. \end{aligned}$$

By (14), $[\nabla_v, L_{\text{anti}}] = -\nabla_x$, hence this term equals $-2\mathcal{C}[h_t]$. Combining the symmetric and anti-symmetric pieces gives

$$\frac{d}{dt} \mathcal{B}[h_t] = -2\gamma \mathcal{B}[h_t] - 2\mathcal{C}[h_t] - 2\gamma T^2 \int h_t \|\nabla_v^2 \log h_t\|_{\text{HS}}^2 d\rho_*,$$

and (17) follows by discarding the final non-positive term.

We now turn to

$$\mathcal{C}[h_t] = T \int h_t \langle \nabla_x \ell_t, \nabla_v \ell_t \rangle d\rho_*.$$

Differentiating and grouping terms gives

$$\begin{aligned} \frac{d}{dt} \mathcal{C}[h_t] &= T \int L_0 h_t \langle \nabla_x \ell_t, \nabla_v \ell_t \rangle d\rho_* \\ &\quad + T \int h_t \langle \nabla_x (\partial_t \ell_t), \nabla_v \ell_t \rangle d\rho_* \\ &\quad + T \int h_t \langle \nabla_x \ell_t, \nabla_v (\partial_t \ell_t) \rangle d\rho_*. \end{aligned}$$

Proceed exactly as for $\mathcal{B}$: rewrite $\partial_t \ell_t = h_t^{-1} L_0 h_t$, integrate by parts, and use the symmetry of L_{sym} and anti-symmetry of L_{anti} . All second-order terms from L_{sym} contribute a remainder $\mathcal{R}_C(t) \leq 0$. The commutator terms are the only first-order contributions, so

$$\begin{aligned} \frac{d}{dt} \mathcal{C}[h_t] &\leq T \int h_t \langle \nabla_x \ell_t, [\nabla_v, L_0] \ell_t \rangle d\rho_* \\ &\quad + T \int h_t \langle \nabla_v \ell_t, [\nabla_x, L_0] \ell_t \rangle d\rho_*. \end{aligned}$$

Applying (14)–(15) gives

$$\frac{d}{dt} \mathcal{C}[h_t] \leq -\mathcal{D}[h_t] - \gamma \mathcal{C}[h_t] + T \int h_t \langle \nabla_x \log h_t, (\nabla^2 U) \nabla_v \log h_t \rangle d\rho_*.$$

Using the operator bound on $\nabla^2 U$ and Cauchy-Schwarz,

$$\left| T \int h_t \langle \nabla_x \log h_t, (\nabla^2 U) \nabla_v \log h_t \rangle d\rho_* \right| \leq M \sqrt{\mathcal{B}[h_t] \mathcal{D}[h_t]},$$

which gives (18).

Finally, starting from

$$\mathcal{D}[h_t] = T \int h_t |\nabla_x \ell_t|^2 d\rho_*,$$

the same differentiation yields

$$\frac{d}{dt} \mathcal{D}[h_t] = 2T \int h_t \langle \nabla_x \log h_t, [\nabla_x, L_0] \log h_t \rangle d\rho_* + \mathcal{R}_D(t),$$

with $\mathcal{R}_D(t) \leq 0$ from the symmetric part, because L_{sym} acts only in velocity. Invoking (15) and Cauchy-Schwarz yields (19). $\square$

Theorem 4.5 (Detailed backbone hypocoercive decay). *Let $M = L_U$ and define*

$$\varepsilon := \min \left\{ 1, \frac{\gamma}{16(M + \gamma)^2} \right\}, \quad a := \frac{\gamma\varepsilon}{4}, \quad b := \varepsilon, \quad c := \varepsilon^2.$$

For the modified entropy

$$\Phi[h] := \mathcal{A}[h] + a\mathcal{B}[h] + 2b\mathcal{C}[h] + c\mathcal{D}[h], \quad (20)$$

there exists $\lambda_{\text{back}} > 0$, depending only on (γ, M) , such that

$$\frac{d}{dt} \Phi[h_t] \leq -\lambda_{\text{back}} (\mathcal{B}[h_t] + \mathcal{D}[h_t]).$$

If the Gibbs measure ρ_ satisfies an LSI with constant $\rho_{\text{LSI}} > 0$, that is,*

$$\rho_{\text{LSI}} \mathcal{A}[h] \leq \mathcal{B}[h] + \mathcal{D}[h],$$

then

$$\Phi[h_t] \leq e^{-\lambda_{\text{back}} \rho_{\text{LSI}} t} \Phi[h_0].$$

Proof. Insert (16)–(19) into the derivative of (20). This gives

$$\begin{aligned} \frac{d}{dt} \Phi[h_t] &\leq -\gamma\mathcal{B} + a(-2\gamma\mathcal{B} - 2\mathcal{C}) + 2b(-\mathcal{D} - \gamma\mathcal{C} + M\sqrt{\mathcal{B}\mathcal{D}}) \\ &\quad + c(2M\sqrt{\mathcal{B}\mathcal{D}}). \end{aligned}$$

Using $-\mathcal{C} \leq \sqrt{\mathcal{B}\mathcal{D}}$, we obtain

$$\frac{d}{dt} \Phi[h_t] \leq -\gamma(1 + 2a)\mathcal{B} - 2b\mathcal{D} + \Xi\sqrt{\mathcal{B}\mathcal{D}},$$

where

$$\Xi := 2a + 2b\gamma + 2M(b + c).$$

Set $Y = (\sqrt{\mathcal{B}}, \sqrt{\mathcal{D}})^\top$. The right-hand side is $-Y^\top KY$ for

$$K = \begin{pmatrix} \gamma(1 + 2a) & -\Xi/2 \\ -\Xi/2 & 2b \end{pmatrix}.$$

Since a, b, c are positive, $K_{11} \geq \gamma$ and $K_{22} = 2\varepsilon$. Moreover,

$$\Xi \leq \frac{\gamma^2\varepsilon}{2} + 2\gamma\varepsilon + 2(M + \gamma)\varepsilon \leq 4(M + \gamma)\varepsilon,$$

because $\varepsilon \leq 1$. Hence

$$\det K \geq 2\gamma\varepsilon - 4(M + \gamma)^2\varepsilon^2 \geq \gamma\varepsilon > 0$$

by the choice of ε . Therefore K is positive definite and

$$Y^\top KY \geq \lambda_{\min}(K)(\mathcal{B} + \mathcal{D})$$

where

$$\lambda_{\min}(K) = \frac{\gamma(1 + 2a) + 2b - \sqrt{(\gamma(1 + 2a) - 2b)^2 + \Xi^2}}{2}.$$

with

$$\lambda_{\text{back}} := \lambda_{\min}(K) > 0.$$

This proves the first claim. If ρ_* satisfies the stated LSI, then

$$\mathcal{B}[h_t] + \mathcal{D}[h_t] \geq \rho_{\text{LSI}}\mathcal{A}[h_t].$$

Since Φ is equivalent to $\mathcal{A} + \mathcal{B} + \mathcal{D}$ for fixed (a, b, c) , the differential inequality closes and Grönwall's lemma gives the exponential decay of Φ . $\square$

Proposition 4.6 (Gibbs logarithmic Sobolev inequality). *Let*

$$W(x, v) := U(x) + \frac{1}{2}\|v\|^2, \quad m_* := \min\{m_U, 1\}.$$

Then the Gibbs measure ρ_ from (7) satisfies*

$$\text{Ent}_{\rho_*}(h) \leq \frac{1}{2m_*}(\mathcal{B}[h] + \mathcal{D}[h]) \quad (21)$$

for every smooth density $h \geq 0$ with $\int h d\rho_ = 1$.*

Proof. The density of ρ_* has the form

$$\rho_*(x, v) = Z_*^{-1} e^{-W(x, v)/T}.$$

The Hessian of W is block diagonal:

$$\nabla^2 W(x, v) = \begin{pmatrix} \nabla^2 U(x) & 0 \\ 0 & I_d \end{pmatrix}.$$

Assumption 2.1, item *ii*, therefore implies

$$\nabla^2 W(x, v) \succeq m_* I_{2d} \quad \forall (x, v) \in \mathbb{R}^{2d}.$$

The Bakry-Émery criterion applied to the probability measure $Z_*^{-1} e^{-W/T} dx dv$ yields

$$\text{Ent}_{\rho_*}(g^2) \leq \frac{2T}{m_*} \int |\nabla g|^2 d\rho_*$$

for every smooth g . Choose $g = \sqrt{h}$. Since

$$|\nabla \sqrt{h}|^2 = \frac{|\nabla_x h|^2 + |\nabla_v h|^2}{4h},$$

we obtain

$$\text{Ent}_{\rho_*}(h) \leq \frac{T}{2m_*} \int \frac{|\nabla_x h|^2 + |\nabla_v h|^2}{h} d\rho_* = \frac{1}{2m_*}(\mathcal{B}[h] + \mathcal{D}[h]),$$

which is (21). $\square$

Lemma 4.7 (Equivalence of the modified entropy). *Let Φ be the modified entropy functional from (20), with the coefficients a, b, c from Theorem 4.5. Then there exists a constant $C_\Phi < \infty$, depending only on $(m_U, L_U, \gamma, \sigma)$, such that*

$$\mathcal{A}[h] \leq \Phi[h] \leq C_\Phi (\mathcal{B}[h] + \mathcal{D}[h])$$

for every smooth density h with $\int h d\pi_1 = 1$.

Proof. Proposition 4.6 yields

$$\mathcal{A}[h] \leq \frac{1}{2m_*} (\mathcal{B}[h] + \mathcal{D}[h]).$$

By Cauchy–Schwarz,

$$2|\mathcal{C}[h]| \leq \mathcal{B}[h] + \mathcal{D}[h].$$

Since

$$\Phi[h] = \mathcal{A}[h] + a\mathcal{B}[h] + 2b\mathcal{C}[h] + c\mathcal{D}[h],$$

we obtain

$$\Phi[h] \leq \left(\frac{1}{2m_*} + a + b + c \right) (\mathcal{B}[h] + \mathcal{D}[h]).$$

The lower bound $\mathcal{A}[h] \leq \Phi[h]$ is immediate from the definition of Φ and the positivity of the quadratic form chosen in Theorem 4.5. $\square$

Corollary 4.8 (One-particle backbone entropy contraction). *Let $P_t^{(1),0}$ be the semigroup of the uncoupled one-particle kinetic backbone and let $\pi_1 = \rho_*$. There exists a constant $\lambda_{\text{hypo}} > 0$, depending only on $(m_U, L_U, \gamma, \sigma)$, such that*

$$\text{KL}(\mu P_t^{(1),0} \parallel \pi_1) \leq e^{-2\lambda_{\text{hypo}} t} \text{KL}(\mu \parallel \pi_1)$$

for every probability measure $\mu \ll \pi_1$ and every $t \geq 0$.

Proof. Let $h_t = d(\mu P_t^{(1),0})/d\pi_1$. By Lemma 4.7,

$$\mathcal{A}[h_t] \leq \Phi[h_t] \quad \text{and} \quad \Phi[h_t] \leq C_\Phi (\mathcal{B}[h_t] + \mathcal{D}[h_t]).$$

Theorem 4.5 gives

$$\frac{d}{dt} \Phi[h_t] \leq -\lambda_{\text{back}} (\mathcal{B}[h_t] + \mathcal{D}[h_t]) \leq -\frac{\lambda_{\text{back}}}{C_\Phi} \Phi[h_t].$$

Hence

$$\Phi[h_t] \leq e^{-(\lambda_{\text{back}}/C_\Phi)t} \Phi[h_0].$$

Applying the same differential inequality from time s onward and integrating from s to ∞ gives

$$\mathcal{A}[h_s] = \gamma \int_s^\infty \mathcal{B}[h_t] dt \leq \gamma \int_s^\infty (\mathcal{B}[h_t] + \mathcal{D}[h_t]) dt \leq \frac{\gamma C_\Phi}{\lambda_{\text{back}}} \Phi[h_s].$$

Using $\mathcal{A}[h_s] \leq \Phi[h_s]$ once more and absorbing constants yields an exponential bound

$$\mathcal{A}[h_t] \leq e^{-2\lambda_{\text{hypo}} t} \mathcal{A}[h_0]$$

for some $\lambda_{\text{hypo}} > 0$ depending only on $(m_U, L_U, \gamma, \sigma)$. This is the stated entropy contraction. $\square$

4.2 Matrix analysis for the viscous velocity block

Definition 4.9 (Viscous matrices). For $X = (x_1, \dots, x_N)$ define

$$K_{ij}(X) := \begin{cases} K(x_i, x_j), & i \neq j, \\ 0, & i = j, \end{cases} \quad D(X) := \text{diag}(d_1(X), \dots, d_N(X)),$$

and

$$W(X) := D(X)^{-1}K(X), \quad L(X) := I - W(X).$$

The velocity dynamics may then be written as

$$dV = (-\nabla U(X) - A(X)V) dt + \sigma dW, \quad A(X) := \gamma I_{Nd} + \nu_{\text{visc}} L(X) \otimes I_d.$$

Lemma 4.10 (Weighted symmetry and spectral bounds). *For every configuration X :*

(i) $D(X)W(X) = K(X)$ is symmetric;

(ii) the matrix

$$\tilde{L}(X) := D(X)^{1/2}L(X)D(X)^{-1/2} = D(X)^{-1/2}(D(X) - K(X))D(X)^{-1/2}$$

is symmetric positive semidefinite;

(iii) every eigenvalue of $\tilde{L}(X)$ lies in $[0, 2]$;

(iv) $A(X)$ is similar to the symmetric positive definite matrix

$$\tilde{A}(X) := \gamma I_{Nd} + \nu_{\text{visc}} \tilde{L}(X) \otimes I_d,$$

hence every eigenvalue of $A(X)$ lies in $[\gamma, \gamma + 2\nu_{\text{visc}}]$.

Proof. The identity $D(X)W(X) = K(X)$ is immediate from the definitions, and $K(X)$ is symmetric because the kernel is symmetric. Therefore

$$\tilde{L}(X) = D(X)^{-1/2}(D(X) - K(X))D(X)^{-1/2}$$

is symmetric. For any $z \in \mathbb{R}^N$,

$$z^\top (D - K)z = \frac{1}{2} \sum_{i \neq j} K_{ij}(z_i - z_j)^2 \geq 0,$$

which proves positive semidefiniteness.

To bound the top of the spectrum, let $y \in \mathbb{R}^N$ and set $z = D(X)^{-1/2}y$. Then

$$y^\top \tilde{L}(X)y = z^\top (D - K)z = \frac{1}{2} \sum_{i \neq j} K_{ij}(z_i - z_j)^2.$$

Also,

$$y^\top y = \sum_i d_i z_i^2.$$

Using $(z_i - z_j)^2 \leq 2z_i^2 + 2z_j^2$ and the symmetry of K ,

$$y^\top \tilde{L}(X)y \leq \sum_{i \neq j} K_{ij}(z_i^2 + z_j^2) = 2 \sum_i d_i z_i^2 = 2 y^\top y.$$

Since $\tilde{L}(X)$ is symmetric positive semidefinite, its Rayleigh quotient belongs to $[0, 2]$, and hence every eigenvalue lies in $[0, 2]$.

Finally,

$$A(X) = (D(X)^{-1/2} \otimes I_d) \tilde{A}(X) (D(X)^{1/2} \otimes I_d),$$

so $A(X)$ and $\tilde{A}(X)$ are similar and hence share the same spectrum. Since $\tilde{L}(X)$ is symmetric positive semidefinite with spectrum in $[0, 2]$, the spectrum of $\tilde{A}(X)$ is contained in $[\gamma, \gamma + 2\nu_{\text{visc}}]$. $\square$

Proposition 4.11 (Frozen-position Gaussian law and covariance bound). *Fix a configuration X and consider the frozen-position velocity process*

$$dV_t = (-b(X) - A(X)V_t) dt + \sigma dW_t, \quad b(X) := (\nabla U(x_1), \dots, \nabla U(x_N)).$$

Then:

(i) *the stationary law is Gaussian*

$$\mathcal{N}(m_X, \Sigma_X), \quad m_X = -A(X)^{-1}b(X);$$

(ii) *the covariance solves the Lyapunov equation*

$$A(X)\Sigma_X + \Sigma_X A(X)^\top = \sigma^2 I_{Nd}; \tag{22}$$

(iii) *its largest eigenvalue satisfies the N -uniform bound*

$$\lambda_{\max}(\Sigma_X) \leq \frac{k^* \sigma^2}{k_* 2\gamma}.$$

Proof. For fixed X , the velocity equation is a linear Ornstein-Uhlenbeck process with constant drift matrix $A(X)$ and constant forcing $-b(X)$. By Lemma 4.10, item *iv*, the spectrum of $A(X)$ lies in $[\gamma, \infty)$, so the linear system is exponentially stable. The unique stationary law is therefore Gaussian with mean

$$m_X = -A(X)^{-1}b(X),$$

and the covariance Σ_X is the unique positive semidefinite solution of (22).

To bound the covariance, write $H(X) := D(X)^{1/2} \otimes I_d$ and

$$A(X) = H(X)^{-1} \tilde{A}(X) H(X), \quad \tilde{A}(X) = \gamma I_{Nd} + \nu_{\text{visc}} \tilde{L}(X) \otimes I_d.$$

Since $\tilde{A}(X)$ is symmetric positive definite, the covariance admits the standard OU representation

$$\Sigma_X = \sigma^2 \int_0^\infty e^{-A(X)t} e^{-A(X)^\top t} dt.$$

Substituting the similarity transform gives

$$\Sigma_X = \sigma^2 H^{-1} \left[\int_0^\infty e^{-\tilde{A}t} H^2 e^{-\tilde{A}t} dt \right] H^{-1}.$$

For any $u \in \mathbb{R}^{Nd}$ with $\|u\| = 1$, set $w = H^{-1}u$. Then

$$u^\top \Sigma_X u = \sigma^2 \int_0^\infty \langle H e^{-\tilde{A}t} w, H e^{-\tilde{A}t} w \rangle dt$$

$$\begin{aligned}
&\leq \sigma^2 \lambda_{\max}(H^2) \int_0^\infty \|e^{-\tilde{A}t} w\|^2 dt \\
&\leq \sigma^2 \lambda_{\max}(H^2) \int_0^\infty e^{-2\gamma t} dt \|w\|^2 \\
&\leq \frac{\sigma^2 \lambda_{\max}(H^2)}{2\gamma \lambda_{\min}(H^2)}.
\end{aligned}$$

Because

$$(N-1)k_* \leq d_i(X) \leq (N-1)k^* \quad \forall i,$$

we have

$$\frac{\lambda_{\max}(H^2)}{\lambda_{\min}(H^2)} = \frac{\max_i d_i(X)}{\min_i d_i(X)} \leq \frac{k^*}{k_*}.$$

Taking the supremum over unit vectors u proves the stated bound on $\lambda_{\max}(\Sigma_X)$. $\square$

Corollary 4.12 (Conditional Gaussian Poincaré inequality). *For every frozen configuration X and every smooth $g : \mathbb{R}^{Nd} \rightarrow \mathbb{R}$,*

$$\text{Var}_{\mathcal{N}(m_X, \Sigma_X)}(g) \leq \frac{k^* \sigma^2}{k_* 2\gamma} \int |\nabla_v g|^2 d\mathcal{N}(m_X, \Sigma_X).$$

Proof. For a centered Gaussian law $\mathcal{N}(m, \Sigma)$, the sharp Poincaré constant is $\lambda_{\max}(\Sigma)$. Since translating by the mean does not change the variance or the gradient term, the same constant applies to $\mathcal{N}(m_X, \Sigma_X)$. Proposition 4.11 bounds $\lambda_{\max}(\Sigma_X)$ by $\frac{k^* \sigma^2}{k_* 2\gamma}$, so the stated inequality follows. $\square$

Lemma 4.13 (Derivative bounds for the normalized weights). *Let*

$$L_K := \sup_{x, y \in \mathbb{R}^d} \|\nabla_x K(x, y)\| < \infty.$$

Then for every $N \geq 2$ and every $i \neq j$,

$$\|\nabla_{x_i} \omega_{ij}(X)\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^* L_K}{(N-1)k_*^2},$$

and

$$\|\nabla_{x_j} \omega_{ij}(X)\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^* L_K}{(N-1)^2 k_*^2},$$

and for $m \notin \{i, j\}$,

$$\|\nabla_{x_m} \omega_{ij}(X)\| \leq \frac{k^* L_K}{(N-1)^2 k_*^2}.$$

In particular,

$$\sup_{N \geq 2} \sup_X \max_{i=1, \dots, N} \max_{m=1, \dots, N} \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\| < \infty.$$

Proof. Differentiate

$$\omega_{ij}(X) = \frac{K(x_i, x_j)}{d_i(X)}, \quad d_i(X) = \sum_{k \neq i} K(x_i, x_k).$$

For the x_i derivative,

$$\nabla_{x_i} \omega_{ij} = \frac{\nabla_{x_i} K(x_i, x_j)}{d_i} - \frac{K(x_i, x_j)}{d_i^2} \sum_{k \neq i} \nabla_{x_i} K(x_i, x_k).$$

Taking norms and using $d_i \geq (N-1)k_*$, $K(x_i, x_j) \leq k^*$, and $\|\nabla_{x_i} K\| \leq L_K$ yields

$$\|\nabla_{x_i} \omega_{ij}\| \leq \frac{L_K}{(N-1)k_*} + \frac{k^*(N-1)L_K}{(N-1)^2 k_*^2}.$$

This is the first estimate. For $m \notin \{i, j\}$ only the denominator depends on x_m , so

$$\nabla_{x_m} \omega_{ij} = -\frac{K(x_i, x_j)}{d_i^2} \nabla_{x_m} K(x_i, x_m),$$

which gives the third estimate. For the x_j derivative,

$$\nabla_{x_j} \omega_{ij} = \frac{\nabla_{x_j} K(x_i, x_j)}{d_i} - \frac{K(x_i, x_j)}{d_i^2} \nabla_{x_j} K(x_i, x_j),$$

and the same lower bound on d_i gives the second estimate. The uniform summability follows by summing the displayed bounds over j for fixed (i, m) . $\square$

Lemma 4.14 (Derivative bounds for the viscous force). *There exists a constant $C_{\partial F} < \infty$, depending only on $(\nu_{\text{visc}}, k_*, k^*, L_K)$, such that for every configuration $(X, V) \in (\mathbb{R}^{2d})^N$:*

$$\max_{1 \leq i \leq N} \|\nabla_V F_i(X, V)\|_{\text{op}} \leq 2\nu_{\text{visc}}, \quad (23)$$

$$\sum_{m=1}^N \|\nabla_{x_m} F_i(X, V)\|^2 \leq C_{\partial F} \left(\|V_i\|^2 + \frac{1}{N} \sum_{j=1}^N \|V_j\|^2 \right). \quad (24)$$

Proof. The velocity derivatives are explicit:

$$\nabla_{v_i} F_i = -\nu_{\text{visc}} I_d, \quad \nabla_{v_m} F_i = \nu_{\text{visc}} \omega_{im}(X) I_d \quad (m \neq i).$$

Hence

$$\|\nabla_V F_i\|_{\text{op}} \leq \nu_{\text{visc}} + \nu_{\text{visc}} \sum_{m \neq i} \omega_{im}(X) = 2\nu_{\text{visc}},$$

which proves (23).

For the x derivatives,

$$\nabla_{x_m} F_i(X, V) = \nu_{\text{visc}} \sum_{j \neq i} \nabla_{x_m} \omega_{ij}(X) (V_j - V_i).$$

Define

$$A_{im}(X) := \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\|.$$

By Lemma 4.13,

$$\sup_{N \geq 2} \sup_X \max_{i, m} A_{im}(X) \leq C_\omega$$

for some finite constant C_ω . Using Cauchy-Schwarz in the discrete sum,

$$\begin{aligned}\|\nabla_{x_m} F_i(X, V)\|^2 &\leq \nu_{\text{visc}}^2 \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\| \right) \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\| \|V_j - V_i\|^2 \right) \\ &\leq \nu_{\text{visc}}^2 C_\omega \left(\sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}\| (2\|V_j\|^2 + 2\|V_i\|^2) \right).\end{aligned}$$

Summing over m , and using the explicit derivative bounds from Lemma 4.13, we have for every $j \neq i$

$$\sum_{m=1}^N \|\nabla_{x_m} \omega_{ij}(X)\| \leq \frac{C_\omega}{N},$$

because only the terms $m \in \{i, j\}$ are of order $(N-1)^{-1}$ while the remaining $(N-2)$ terms are of order $(N-1)^{-2}$. Summing over $j \neq i$, this yields

$$\sum_{m=1}^N \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\| \leq C_\omega.$$

Then, by the same bound,

$$\sum_{m=1}^N \sum_{j \neq i} \|\nabla_{x_m} \omega_{ij}(X)\| \|V_j\|^2 \leq C_\omega \frac{1}{N} \sum_{j=1}^N \|V_j\|^2.$$

Collecting terms yields (24). □

5 Backbone Tensorization

5.1 Tensorized backbone gap

This section upgrades the one-particle hypocoercive estimate from Section 4 to the product reference structure used later in both the finite-time and stationary arguments. It is preparatory: the tensorized backbone provides the canonical uncoupled comparison object, while the main mean-field results are proved in the sections that follow.

For the kinetic backbone ($\nu_{\text{visc}} = 0$), the invariant law is $\pi_1^{\otimes N}$ and the gap from Corollary 4.8 tensorizes immediately.

Proposition 5.1 (Tensorized backbone entropy gap). *Let $P_t^{N,0}$ denote the semigroup of the uncoupled N -particle kinetic backbone obtained by setting $\nu_{\text{visc}} = 0$ in Definition 2.5. Then*

$$\text{KL}(\mu P_t^{N,0} \parallel \pi_1^{\otimes N}) \leq e^{-2\lambda_{\text{hypo}} t} \text{KL}(\mu \parallel \pi_1^{\otimes N})$$

for every $\mu \ll \pi_1^{\otimes N}$ and every $N \geq 1$.

Proof. The uncoupled generator is the sum of the N one-particle generators, so the semigroup factorizes as

$$P_t^{N,0} = (P_t^{(1),0})^{\otimes N}.$$

Let $\mu \ll \pi_1^{\otimes N}$ and write

$$h_0 = \frac{d\mu}{d\pi_1^{\otimes N}}.$$

Let $h_t := \frac{d(\mu P_t^{N,0})}{d\pi_1^{\otimes N}}$. The relative entropy with respect to a product reference measure satisfies the usual conditional chain rule:

$$\text{KL}(\mu P_t^{N,0} \parallel \pi_1^{\otimes N}) = \sum_{i=1}^N \mathbb{E}_{\mu P_t^{N,0}} \left[\text{KL}((\mu P_t^{N,0})_{i|1:i-1}(\cdot \mid Z_1, \dots, Z_{i-1}) \parallel \pi_1) \right].$$

Because $P_t^{N,0}$ acts independently on each coordinate and preserves π_1 , the i -th conditional law evolves by the one-particle semigroup $P_t^{(1),0}$ while the earlier coordinates are frozen. Corollary 4.8 therefore gives, for every conditioning value,

$$\text{KL}((\mu P_t^{N,0})_{i|1:i-1}(\cdot \mid Z_1, \dots, Z_{i-1}) \parallel \pi_1) \leq e^{-2\lambda_{\text{hypo}} t} \text{KL}(\mu_{i|1:i-1}(\cdot \mid Z_1, \dots, Z_{i-1}) \parallel \pi_1).$$

Taking expectations and summing over i gives

$$\text{KL}(\mu P_t^{N,0} \parallel \pi_1^{\otimes N}) \leq e^{-2\lambda_{\text{hypo}} t} \text{KL}(\mu \parallel \pi_1^{\otimes N})$$

for every $\mu \ll \pi_1^{\otimes N}$ and every $N \geq 1$. □

Remark 5.2 (Hypocoercive carré du champ). For the backbone dynamics one may choose a Villani-type quadratic form

$$\Gamma_{\text{hypo}}^N(g, g) := a \|\nabla_v g\|^2 + 2b \nabla_x g \cdot \nabla_v g + c \|\nabla_x g\|^2, \quad ac > b^2,$$

with coefficients depending only on (U, γ, σ) and not on N , so that Γ_{hypo}^N is equivalent to the full H^1 norm under the backbone invariant law. The entropy gap of Proposition 5.1 is equivalent to an estimate of the form

$$\text{Ent}_{\pi_1^{\otimes N}}(g^2) \leq C_{\text{hypo}} \int \Gamma_{\text{hypo}}^N(g, g) d\pi_1^{\otimes N}.$$

We now explain why the viscous alignment does not destroy this structure; the same hypocoercive coefficients will be reused later in the stationary mean-field logarithmic Sobolev inequality, whose modified-entropy mechanism goes back to [?].

The remainder of the paper is devoted to the mean-field stationary theory: finite-particle Harris estimates and stationary compactness, identification of the stationary McKean–Vlasov limit, and a stationary mean-field logarithmic Sobolev inequality in fixed dimension.

6 Finite-Time Mean-Field Approximation

6.1 The McKean–Vlasov limit

We now compare the finite-particle law to the nonlinear mean-field equation (5). The key fact is that the interaction field from Definition 2.6 is globally Lipschitz in both the state variables and the underlying measure, thanks to the strictly positive kernel floor.

Lemma 6.1 (Empirical-force defect). *Let*

$$\mu_t^N := \frac{1}{N} \sum_{j=1}^N \delta_{(X_j(t), V_j(t))}$$

be the empirical measure of the particle system and, for each i , let

$$F_i^N(t) := \nu_{\text{visc}} \sum_{j \neq i} \omega_{ij}(X(t))(V_j(t) - V_i(t))$$

denote the exact finite-particle alignment force. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N , such that

$$\sup_{t \in [0, T]} \frac{1}{N} \sum_{i=1}^N \mathbb{E} \left[\|F_i^N(t) - F[\mu_t^N](X_i(t), V_i(t))\|^2 \right] \leq \frac{C_T}{N}.$$

Remark 6.2 (Where the normalization matters). The state-dependent denominator is the main model-specific issue in the finite-time mean-field analysis. The strict kernel floor $k_* > 0$ prevents any singularity in the normalization, while the self-interaction defect enters only through the difference between $\sum_{k \neq i} K(X_i, X_k)$ and $\sum_{k=1}^N K(X_i, X_k)$. This is exactly why the normalized interaction still produces an $O(N^{-1})$ force defect and remains compatible with the quantitative propagation-of-chaos bounds proved below.

Proof. Fix i and abbreviate $X_i = X_i(t)$, $V_i = V_i(t)$. By definition,

$$F[\mu_t^N](X_i, V_i) = \frac{\nu_{\text{visc}}}{m_{\mu_t^N}(X_i)} \frac{1}{N} \sum_{j=1}^N K(X_i, X_j)(V_j - V_i),$$

where

$$m_{\mu_t^N}(X_i) = \frac{1}{N} \sum_{j=1}^N K(X_i, X_j).$$

Since $K(X_i, X_i) = 1 + \kappa_0$, the finite-particle force and the empirical-field force differ only because of the self-interaction term and the normalization:

$$F_i^N - F[\mu_t^N](X_i, V_i) = \nu_{\text{visc}} \left[\sum_{j \neq i} \frac{K(X_i, X_j)}{\sum_{k \neq i} K(X_i, X_k)} (V_j - V_i) - \frac{\sum_{j \neq i} K(X_i, X_j)(V_j - V_i)}{\sum_{k=1}^N K(X_i, X_k)} \right].$$

Write

$$S_i := \sum_{j \neq i} K(X_i, X_j)(V_j - V_i), \quad d_i := \sum_{k \neq i} K(X_i, X_k).$$

Then

$$F_i^N - F[\mu_t^N](X_i, V_i) = \nu_{\text{visc}} S_i \left(\frac{1}{d_i} - \frac{1}{d_i + K(X_i, X_i)} \right) = \nu_{\text{visc}} S_i \frac{K(X_i, X_i)}{d_i(d_i + K(X_i, X_i))}.$$

By Assumption 2.3,

$$d_i \geq (N-1)k_*, \quad K(X_i, X_i) \leq k^*,$$

and therefore

$$\left| \frac{K(X_i, X_i)}{d_i(d_i + K(X_i, X_i))} \right| \leq \frac{k^*}{(N-1)^2 k_*^2}.$$

On the other hand,

$$\|S_i\| \leq \sum_{j \neq i} K(X_i, X_j) \|V_j - V_i\| \leq 2k^* \sum_{j \neq i} (\|V_j\| + \|V_i\|).$$

Using the uniform second-moment bounds on the particle system over finite time intervals, which follow from Proposition 3.6, we obtain

$$\mathbb{E} \|S_i\|^2 \leq C_T N^2$$

with C_T independent of N . Combining the last three displays yields

$$\mathbb{E} \|F_i^N - F[\mu_t^N](X_i, V_i)\|^2 \leq \frac{C_T}{N^2}.$$

Averaging over i proves the claim. $\square$

Lemma 6.3 (Monte Carlo fluctuation of the mean-field force). *Let $(\bar{X}_i(t), \bar{V}_i(t))_{i=1}^N$ be i.i.d. copies of the nonlinear McKean-Vlasov process with common law f_t , and let*

$$\bar{\mu}_t^N := \frac{1}{N} \sum_{j=1}^N \delta_{(\bar{X}_j(t), \bar{V}_j(t))}$$

be the associated empirical measure. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N , such that

$$\sup_{t \in [0, T]} \frac{1}{N} \sum_{i=1}^N \mathbb{E} \left[\|F[\bar{\mu}_t^N](\bar{X}_i(t), \bar{V}_i(t)) - F[f_t](\bar{X}_i(t), \bar{V}_i(t))\|^2 \right] \leq \frac{C_T}{N}.$$

Proof. By exchangeability it suffices to estimate the index $i = 1$. Fix $t \in [0, T]$ and condition on $(\bar{X}_1, \bar{V}_1) = (x, v)$. Define

$$N_N(x, v) := \frac{1}{N} \sum_{j=1}^N K(x, \bar{X}_j)(\bar{V}_j - v), \quad D_N(x) := \frac{1}{N} \sum_{j=1}^N K(x, \bar{X}_j),$$

and also

$$n_t(x, v) := \iint K(x, y)(q - v) f_t(dy, dq), \quad m_t(x) := \int K(x, y) \rho_{f_t}(dy).$$

Then

$$F[\bar{\mu}_t^N](x, v) = \nu_{\text{visc}} \frac{N_N(x, v)}{D_N(x)}, \quad F[f_t](x, v) = \nu_{\text{visc}} \frac{n_t(x, v)}{m_t(x)}.$$

Because $D_N(x) \geq k_*$ and $m_t(x) \geq k_*$,

$$\left\| \frac{N_N}{D_N} - \frac{n_t}{m_t} \right\| \leq \frac{\|N_N - n_t\|}{k_*} + \frac{\|n_t\|}{k_*^2} |D_N - m_t|.$$

Conditionally on $(\bar{X}_1, \bar{V}_1) = (x, v)$, the variables

$$Y_j := K(x, \bar{X}_j)(\bar{V}_j - v), \quad Z_j := K(x, \bar{X}_j)$$

are independent for $j \geq 2$. Their conditional second moments satisfy

$$\mathbb{E}[\|Y_j\|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq 2(k^*)^2(\mathbb{E}\|\bar{V}_j\|^2 + \|v\|^2), \quad \mathbb{E}[|Z_j|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq (k^*)^2.$$

The $j = 1$ contributions to N_N and D_N are deterministic once (x, v) is fixed and are of size $O(N^{-1}(1 + \|v\|))$, so they can be absorbed into the same N^{-1} bounds. Therefore the empirical averages have conditional variances

$$\mathbb{E}[\|N_N(x, v) - n_t(x, v)\|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq \frac{C_T(1 + \|v\|^2)}{N},$$

and

$$\mathbb{E}[|D_N(x) - m_t(x)|^2 \mid \bar{X}_1 = x, \bar{V}_1 = v] \leq \frac{C_T}{N}.$$

Moreover,

$$\|n_t(x, v)\| \leq k^* \int (\|q\| + \|v\|) f_t(dy, dq) \leq C_T(1 + \|v\|).$$

Combining the last three displays gives

$$\mathbb{E}[\|F[\bar{\mu}_t^N](\bar{X}_1, \bar{V}_1) - F[f_t](\bar{X}_1, \bar{V}_1)\|^2 \mid \bar{X}_1, \bar{V}_1] \leq \frac{C_T(1 + \|\bar{V}_1\|^2)}{N}.$$

Integrating over $(\bar{X}_1, \bar{V}_1)$ and using Proposition 3.6 yields the claim. $\square$

Theorem 6.4 (Strong propagation of chaos). *Let $(\bar{X}_i, \bar{V}_i)_{i=1}^N$ be i.i.d. solutions of the nonlinear McKean-Vlasov SDE with initial law f_0 , driven by the same Brownian motions as the particle system (1)–(2), and started from the same initial data:*

$$(X_i(0), V_i(0)) = (\bar{X}_i(0), \bar{V}_i(0)) \quad \text{a.s. for every } i.$$

Set

$$\varepsilon_N(t) := \frac{1}{N} \sum_{i=1}^N \mathbb{E}[\|X_i(t) - \bar{X}_i(t)\|^2 + \|V_i(t) - \bar{V}_i(t)\|^2].$$

Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N , such that

$$\sup_{t \in [0, T]} \varepsilon_N(t) \leq \frac{C_T}{N}.$$

Proof. Write

$$e_i^x := X_i - \bar{X}_i, \quad e_i^v := V_i - \bar{V}_i, \quad \Delta_i := F_i^N(X, V) - F[f_t](\bar{X}_i, \bar{V}_i).$$

Then

$$de_i^x = e_i^v dt, \quad de_i^v = (-\nabla U(X_i) + \nabla U(\bar{X}_i) - \gamma e_i^v + \Delta_i) dt.$$

Itô's formula yields

$$\frac{d}{dt} \mathbb{E}\|e_i^x\|^2 = 2\mathbb{E}\langle e_i^x, e_i^v \rangle \leq \mathbb{E}\|e_i^x\|^2 + \mathbb{E}\|e_i^v\|^2,$$

and, by the Lipschitz bound on ∇U ,

$$\begin{aligned} \frac{d}{dt} \mathbb{E} \|e_i^v\|^2 &= 2\mathbb{E} \langle e_i^v, -\nabla U(X_i) + \nabla U(\bar{X}_i) - \gamma e_i^v + \Delta_i \rangle \\ &\leq (L_U + 1) \mathbb{E} \|e_i^x\|^2 + (L_U + 1 - 2\gamma) \mathbb{E} \|e_i^v\|^2 + \mathbb{E} \|\Delta_i\|^2. \end{aligned}$$

Adding the two inequalities and averaging over i gives

$$\frac{d}{dt} \varepsilon_N(t) \leq C_0 \varepsilon_N(t) + \frac{1}{N} \sum_{i=1}^N \mathbb{E} \|\Delta_i(t)\|^2 \quad (25)$$

for a constant C_0 depending only on (L_U, γ) .

Decompose

$$\Delta_i = R_i + S_i + T_i,$$

where

$$\begin{aligned} R_i &:= F_i^N(X, V) - F[\mu_t^N](X_i, V_i), \\ S_i &:= F[\mu_t^N](X_i, V_i) - F[\bar{\mu}_t^N](\bar{X}_i, \bar{V}_i), \\ T_i &:= F[\bar{\mu}_t^N](\bar{X}_i, \bar{V}_i) - F[f_t](\bar{X}_i, \bar{V}_i). \end{aligned}$$

Using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$,

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E} \|\Delta_i\|^2 \leq 3 \left(\frac{1}{N} \sum_i \mathbb{E} \|R_i\|^2 + \frac{1}{N} \sum_i \mathbb{E} \|S_i\|^2 + \frac{1}{N} \sum_i \mathbb{E} \|T_i\|^2 \right).$$

Lemma 6.1 yields

$$\frac{1}{N} \sum_i \mathbb{E} \|R_i\|^2 \leq \frac{C_T}{N},$$

and Lemma 6.3 yields

$$\frac{1}{N} \sum_i \mathbb{E} \|T_i\|^2 \leq \frac{C_T}{N}.$$

For the middle term, Lemma 3.4 gives

$$\|S_i\| \leq L_{\text{mf}} \left(\|e_i^x\| + \|e_i^v\| + W_1(\mu_t^N, \bar{\mu}_t^N) \right).$$

Coupling the empirical measures particlewise yields

$$W_1(\mu_t^N, \bar{\mu}_t^N) \leq \frac{1}{N} \sum_{j=1}^N (\|e_j^x\| + \|e_j^v\|),$$

and Jensen's inequality therefore gives

$$\mathbb{E} W_1(\mu_t^N, \bar{\mu}_t^N)^2 \leq 2\varepsilon_N(t).$$

Consequently

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E} \|S_i\|^2 \leq C_1 \varepsilon_N(t).$$

Substituting into (25) yields

$$\frac{d}{dt}\varepsilon_N(t) \leq C_2\varepsilon_N(t) + \frac{C_T}{N}.$$

Because $\varepsilon_N(0) = 0$, Grönwall's lemma gives

$$\sup_{t \in [0, T]} \varepsilon_N(t) \leq \frac{C_T}{N}.$$

□

Theorem 6.5 (Relative entropy on path space and at fixed time). *Under the hypotheses of Theorem 6.4, let $P_{[0, T]}^N$ be the law on path space $C([0, T]; (\mathbb{R}^{2d})^N)$ of the interacting system (1)–(2), and let $Q_{[0, T]}^N$ be the law of the independent nonlinear system from Theorem 6.4. Then for every finite horizon $T > 0$ there exists $C_T < \infty$, independent of N , such that*

$$\text{KL}(P_{[0, T]}^N \parallel Q_{[0, T]}^N) \leq C_T. \quad (26)$$

In particular, for every $t \in [0, T]$,

$$\text{KL}(\rho_t^N \parallel f_t^{\otimes N}) \leq C_T. \quad (27)$$

Proof. Under $Q_{[0, T]}^N$, the processes satisfy

$$d\bar{V}_i = (-\nabla U(\bar{X}_i) - \gamma \bar{V}_i + F[f_t](\bar{X}_i, \bar{V}_i)) dt + \sigma dW_i.$$

The interacting system differs only in the velocity drift. Since the diffusion matrix is constant in the velocity variables, Girsanov's theorem applies. The Radon-Nikodým derivative of $P_{[0, T]}^N$ with respect to $Q_{[0, T]}^N$ is

$$\frac{dP_{[0, T]}^N}{dQ_{[0, T]}^N} = \exp\left(\frac{1}{\sigma} \sum_{i=1}^N \int_0^T \Delta_i(s) \cdot dW_i(s) - \frac{1}{2\sigma^2} \sum_{i=1}^N \int_0^T \|\Delta_i(s)\|^2 ds\right),$$

with

$$\Delta_i(s) := F_i^N(X_s, V_s) - F[f_s](X_i(s), V_i(s)).$$

The moment bounds from Proposition 3.6, together with Lemma 6.1 and Theorem 6.4, imply Novikov's criterion on every finite interval. Taking expectations under $P_{[0, T]}^N$ gives the exact entropy identity

$$\text{KL}(P_{[0, T]}^N \parallel Q_{[0, T]}^N) = \frac{1}{2\sigma^2} \sum_{i=1}^N \mathbb{E} \int_0^T \|\Delta_i(s)\|^2 ds.$$

We estimate the drift discrepancy by inserting the coupled empirical measure $\bar{\mu}_t^N$ from Theorem 6.4. Write

$$\Delta_i = R_i + U_i + T_i + V_i,$$

where

$$\begin{aligned} R_i &:= F_i^N(X, V) - F[\mu_t^N](X_i, V_i), \\ U_i &:= F[\mu_t^N](X_i, V_i) - F[\bar{\mu}_t^N](\bar{X}_i, \bar{V}_i), \\ T_i &:= F[\bar{\mu}_t^N](\bar{X}_i, \bar{V}_i) - F[f_t](\bar{X}_i, \bar{V}_i), \\ V_i &:= F[f_t](\bar{X}_i, \bar{V}_i) - F[f_t](X_i, V_i). \end{aligned}$$

Then $(a + b + c + d)^2 \leq 4(a^2 + b^2 + c^2 + d^2)$ gives

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E} \|\Delta_i(s)\|^2 \leq \frac{4}{N} \sum_{i=1}^N \mathbb{E} \|R_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^N \mathbb{E} \|U_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^N \mathbb{E} \|T_i(s)\|^2 + \frac{4}{N} \sum_{i=1}^N \mathbb{E} \|V_i(s)\|^2.$$

Lemma 6.1 controls the R_i term, and Lemma 6.3 controls the T_i term:

$$\frac{1}{N} \sum_i \mathbb{E} \|R_i(s)\|^2 \leq \frac{C_T}{N}, \quad \frac{1}{N} \sum_i \mathbb{E} \|T_i(s)\|^2 \leq \frac{C_T}{N}.$$

For the U_i term, Lemma 3.4 gives

$$\|U_i\| \leq L_{\text{mf}} \left(\|X_i - \bar{X}_i\| + \|V_i - \bar{V}_i\| + W_1(\mu_t^N, \bar{\mu}_t^N) \right).$$

Exactly as in the proof of Theorem 6.4,

$$\mathbb{E} W_1(\mu_t^N, \bar{\mu}_t^N)^2 \leq 2\varepsilon_N(t),$$

and therefore

$$\frac{1}{N} \sum_i \mathbb{E} \|U_i(s)\|^2 \leq C_T \varepsilon_N(s).$$

For the last term, the Lipschitz bound of Lemma 3.4 gives

$$\frac{1}{N} \sum_i \mathbb{E} \|V_i(s)\|^2 \leq C_T \varepsilon_N(s).$$

Combining the last four displays and using Theorem 6.4, we obtain

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E} \|\Delta_i(s)\|^2 \leq C_T \varepsilon_N(s) + \frac{C_T}{N} \leq \frac{C_T}{N}$$

for all $s \in [0, T]$. Integrating in time gives

$$\sum_{i=1}^N \mathbb{E} \int_0^T \|\Delta_i(s)\|^2 ds \leq C_T,$$

which proves (26). The fixed-time bound (27) follows by data processing, since evaluation at time t is a measurable map from path space to $(\mathbb{R}^{2d})^N$. $\square$

Remark 6.6. The path-space entropy bound (26) is extensive but uniform in N : over every finite horizon the full interacting law differs from the independent nonlinear reference law by only $O(1)$ relative entropy on path space. It therefore complements the strong synchronous-coupling estimate of Theorem 6.4 by giving an information-theoretic formulation of the finite-time mean-field limit that automatically controls every fixed-time marginal through data processing.

Corollary 6.7 (One-particle total-variation error). *Let $\rho_{t,1}^N$ be the first marginal of ρ_t^N . Under the assumptions of Theorem 6.5,*

$$\|\rho_{t,1}^N - f_t\|_{\text{TV}} \leq \sqrt{\frac{2C_T}{N}} \quad \forall t \in [0, T].$$

Proof. For any probability measure P^N on $(\mathbb{R}^{2d})^N$ and any product measure $Q^{\otimes N}$, the chain rule for relative entropy gives

$$\text{KL}(P^N \parallel Q^{\otimes N}) = \sum_{i=1}^N \mathbb{E}_{P^N} \left[\text{KL}(P_{i|1:i-1}^N(\cdot \mid Z_1, \dots, Z_{i-1}) \parallel Q) \right].$$

Since $\nu \mapsto \text{KL}(\nu \parallel Q)$ is convex,

$$\mathbb{E}_{P^N} \left[\text{KL}(P_{i|1:i-1}^N(\cdot \mid Z_1, \dots, Z_{i-1}) \parallel Q) \right] \geq \text{KL}(P_i^N \parallel Q).$$

Applying this to $P^N = \rho_t^N$ and $Q = f_t$, and using exchangeability of ρ_t^N , we obtain

$$\text{KL}(\rho_{t,1}^N \parallel f_t) \leq \frac{1}{N} \text{KL}(\rho_t^N \parallel f_t^{\otimes N}).$$

Now use (27) and Pinsker's inequality. □

Corollary 6.8 (Fixed- k marginal convergence on compact time intervals). *Fix $k \geq 1$. Let $\rho_{t,k}^N$ be the k -particle marginal of ρ_t^N . Under the assumptions of Theorem 6.5, for every finite horizon $T > 0$ there exists $C_{T,k} < \infty$, independent of N , such that*

$$\text{KL}(\rho_{t,k}^N \parallel f_t^{\otimes k}) \leq \frac{C_{T,k}}{N}, \quad \|\rho_{t,k}^N - f_t^{\otimes k}\|_{\text{TV}} \leq \sqrt{\frac{2C_{T,k}}{N}} \quad \forall t \in [0, T].$$

One may take $C_{T,k} = k C_T$, where C_T is the constant from Theorem 6.5.

Proof. The same chain-rule argument used in Corollary 6.7 gives, for every exchangeable probability measure P^N on $(\mathbb{R}^{2d})^N$ and every product measure $Q^{\otimes N}$,

$$\text{KL}(P_k^N \parallel Q^{\otimes k}) \leq \sum_{i=1}^k \text{KL}(P_i^N \parallel Q).$$

By exchangeability, each one-particle marginal equals P_1^N , so

$$\text{KL}(P_k^N \parallel Q^{\otimes k}) \leq k \text{KL}(P_1^N \parallel Q) \leq \frac{k}{N} \text{KL}(P^N \parallel Q^{\otimes N}).$$

Apply this with $P^N = \rho_t^N$ and $Q = f_t$, and use (27) to obtain

$$\text{KL}(\rho_{t,k}^N \parallel f_t^{\otimes k}) \leq \frac{k C_T}{N}.$$

Pinsker's inequality then gives the total-variation estimate. □

7 Stationary Thermodynamic Limit and Stationary Functional Inequalities

The stationary analysis proceeds in a fixed order. We first construct the finite- N invariant laws by Harris theory. We then use exchangeability and de Finetti reduction to obtain tightness of the stationary one-particle marginals and identify every subsequential limit as a stationary McKean–Vlasov law. In the perturbative small-viscosity regime we prove uniqueness of that stationary

mean-field law, which upgrades subsequential convergence to the full stationary propagation-of-chaos statement. Finally, we transfer a perturbative logarithmic Sobolev inequality from the frozen linear generator to the unique stationary mean-field law, yielding the perturbative stationary mean-field LSI.

Exchangeability is the bridge from finite-particle invariant laws to de Finetti reduction and stationary thermodynamic identification.

7.1 Finite-particle invariant laws via Harris theory

The stationary arguments below require the finite-particle invariant laws to be constructed inside the present document. We therefore record the same Lyapunov/minorization/Harris package used in Volume 3, now specialized to the continuous kinetic diffusion (1)–(2).

Proposition 7.1 (Compact-set minorization for the finite-particle diffusion). *Fix $N \geq 2$, $t_0 > 0$, and a compact set $K \subset (\mathbb{R}^{2d})^N$. Then there exist $\varepsilon_K > 0$ and a probability measure ν_K on $(\mathbb{R}^{2d})^N$ such that*

$$P_{t_0}^N(z, A) \geq \varepsilon_K \nu_K(A) \quad \forall z \in K$$

for every Borel set $A \subset (\mathbb{R}^{2d})^N$.

Proof. The generator $\mathcal{L}_N$ has smooth coefficients, the velocity noise is non-degenerate in every particle, and the transport field $v \cdot \nabla_x$ propagates that noise to the position variables. Hence the kinetic diffusion satisfies the standard Hörmander bracket condition, so for every $t_0 > 0$ the transition kernel $P_{t_0}^N$ admits a smooth density

$$P_{t_0}^N(z, dz') = p_{t_0}^N(z, z') dz'$$

that is jointly continuous and strictly positive. Choose a Euclidean ball $B \subset (\mathbb{R}^{2d})^N$ with positive Lebesgue measure. Since $K \times B$ is compact and $p_{t_0}^N > 0$ on that set, the minimum

$$\varepsilon_K := |B| \inf_{(z, z') \in K \times B} p_{t_0}^N(z, z')$$

is strictly positive. Let ν_K be normalized Lebesgue measure on B . Then for every Borel A ,

$$P_{t_0}^N(z, A) = \int_A p_{t_0}^N(z, z') dz' \geq \varepsilon_K \nu_K(A) \quad \forall z \in K,$$

which is the required minorization. $\square$

Theorem 7.2 (Lebesgue irreducibility and aperiodicity). *For every $N \geq 2$, the finite-particle kinetic semigroup P_t^N is irreducible with respect to Lebesgue measure and is aperiodic on $(\mathbb{R}^{2d})^N$.*

Proof. Fix $t_0 > 0$. By Proposition 7.1, every compact set is a small set for $P_{t_0}^N$. Because the transition density $p_{t_0}^N(z, z')$ is strictly positive, any non-empty open set has positive $P_{t_0}^N(z, \cdot)$ -mass from every starting point z , which is exactly irreducibility with respect to Lebesgue measure. Choosing a compact ball K and applying Proposition 7.1 with a ball $B \subset K$ shows that $P_{t_0}^N(z, K) > 0$ for all $z \in K$. This one-step return to a small set rules out periodicity, so the semigroup is aperiodic. $\square$

Proposition 7.3 (Uniform kinetic structure for the finite-particle drift). *For every $N \geq 2$, the generator of the finite-particle diffusion can be written as*

$$\mathcal{L}_N \phi = V \cdot \nabla_X \phi + (-\nabla U_N(X) - A_N(X)V) \cdot \nabla_V \phi + \gamma T \Delta_V \phi, \quad T = \frac{\sigma^2}{2\gamma},$$

where

$$U_N(X) := \sum_{i=1}^N U(x_i), \quad A_N(X) := \gamma I_{Nd} + \nu_{\text{visc}} L(X) \otimes I_d.$$

Moreover:

(i) $U_N \in C^2((\mathbb{R}^d)^N)$, $\|\nabla^2 U_N\|_{\text{op}} \leq L_U$, and

$$\langle X, \nabla U_N(X) \rangle \geq \alpha_U \|X\|^2 - N\beta_U;$$

(ii) $A_N(X)$ depends smoothly on X , and

$$\|A_N(X)\|_{\text{op}} \leq \gamma + 2\nu_{\text{visc}};$$

(iii) every eigenvalue of $A_N(X)$ lies in $[\gamma, \gamma + 2\nu_{\text{visc}}]$, uniformly in N and X ;

(iv) the diffusion matrix is constant and non-degenerate in the velocity variables.

Proof. The representation of $\mathcal{L}_N$ follows directly from Definition 2.5 and Definition 4.9. Item *i* is just Assumption 2.1, summed over the particles. Item *ii* follows from Lemma 4.10, because the spectrum of $\tilde{L}(X)$ is contained in $[0, 2]$. Item *iii* is exactly Lemma 4.10(iv). Item *iv* is immediate from the form of the noise term σdW_t . $\square$

Theorem 7.4 (Finite-particle Harris theorem). *For every $N \geq 2$, the diffusion (1)–(2) admits a unique invariant law π_N . Moreover there exists a function $V_N : (\mathbb{R}^{2d})^N \rightarrow [1, \infty)$, constants $c_-, c_+, \lambda_V, C_V > 0$, depending only on $(\alpha_U, \beta_U, L_U, \gamma, \sigma, \nu_{\text{visc}}, k_*, k^*)$ and independent of N , such that*

$$V_N(X, V) \asymp 1 + \frac{1}{N} \sum_{i=1}^N (\|x_i\|^2 + \|v_i\|^2), \quad (28)$$

$$\mathcal{L}_N V_N \leq -\lambda_V V_N + C_V. \quad (29)$$

Consequently,

$$\int V_N d\pi_N \leq \frac{C_V}{\lambda_V}. \quad (30)$$

Proof. We verify the exact structural hypotheses used in the classical Harris theory for underdamped Langevin diffusions.

Step 1: kinetic-Langevin structure. Proposition 7.3 shows that the generator has the standard kinetic form with uniformly convex confining potential U_N , constant non-degenerate velocity noise, and an affine velocity drift $-A_N(X)V$ whose spectrum stays in the fixed interval $[\gamma, \gamma + 2\nu_{\text{visc}}]$. In particular, the only constants entering the classical Lyapunov construction are $(\alpha_U, \beta_U, L_U, \gamma, \sigma, \nu_{\text{visc}}, k_*, k^*)$, and all of them are independent of N .

Step 2: non-explosion and smooth semigroup. Proposition 3.5 gives a unique global strong solution for every initial condition. Since the diffusion matrix is constant in the velocity variables and the coefficients are smooth, the semigroup is the standard smooth kinetic semigroup covered by the Roberts-Tweedie/Meyn-Tweedie Harris framework for underdamped diffusions.

Step 3: Lyapunov function. For this kinetic class, the classical proof constructs a quadratic cross-term Lyapunov function of the form

$$\tilde{V}_N(X, V) := 1 + \eta_1 \frac{1}{N} U_N(X) + \eta_2 \frac{1}{N} \|V\|^2 + \eta_3 \frac{1}{N} \langle X, V \rangle,$$

with coefficients $\eta_1, \eta_2, \eta_3 > 0$ depending only on the structural constants from Step 1. The proof uses only the coercive bound on $\langle X, \nabla U_N(X) \rangle$, the Hessian bound $\|\nabla^2 U_N\|_{\text{op}} \leq L_U$, the uniform spectral window for $A_N(X)$, and the non-degenerate velocity diffusion. Because all of those bounds are uniform in N , the same coefficient choice gives constants $c_-, c_+, \lambda_V, C_V > 0$, independent of N , such that

$$c_- \left(1 + \frac{1}{N} \sum_{i=1}^N (\|x_i\|^2 + \|v_i\|^2) \right) \leq \tilde{V}_N(X, V) \leq c_+ \left(1 + \frac{1}{N} \sum_{i=1}^N (\|x_i\|^2 + \|v_i\|^2) \right)$$

and

$$\mathcal{L}_N \tilde{V}_N \leq -\lambda_V \tilde{V}_N + C_V.$$

This is exactly (28)–(29), after renaming $\tilde{V}_N$ as V_N .

Step 4: Harris theorem and invariant-law bound. Proposition 7.1 and Theorem 7.2 provide the small-set, irreducibility, and aperiodicity hypotheses. The Harris theorem [?, ?] therefore yields existence and uniqueness of the invariant probability law π_N and geometric ergodicity of P_t^N . Integrating (29) against π_N gives (30). $\square$

7.2 Exchangeability and finite de Finetti reduction

Theorem 7.5 (Exchangeability of the invariant law). *For every $N \geq 2$, the invariant law π_N from Theorem 7.4 is exchangeable:*

$$\sigma_{\#} \pi_N = \pi_N \quad \forall \sigma \in S_N,$$

where σ acts on $(\mathbb{R}^{2d})^N$ by permutation of particle labels.

Proof. Fix a permutation $\sigma \in S_N$ and let

$$\Sigma_{\sigma}(x_1, v_1, \dots, x_N, v_N) := (x_{\sigma(1)}, v_{\sigma(1)}, \dots, x_{\sigma(N)}, v_{\sigma(N)}).$$

Because the confining force $-\nabla U(x_i)$, the friction $-\gamma v_i$, the velocity noise σdW_i , and the row-normalized viscous force

$$F_i(X, V) = \nu_{\text{visc}} \sum_{j \neq i} \omega_{ij}(X)(v_j - v_i)$$

are defined by the same formula for every label i , the generator $\mathcal{L}_N$ of (1)–(2) commutes with every permutation:

$$\mathcal{L}_N(\varphi \circ \Sigma_{\sigma}) = (\mathcal{L}_N \varphi) \circ \Sigma_{\sigma} \quad \forall \varphi \in C_c^{\infty}((\mathbb{R}^{2d})^N).$$

Equivalently, if P_t^N is the Markov semigroup of the particle system, then

$$P_t^N(\varphi \circ \Sigma_{\sigma}) = (P_t^N \varphi) \circ \Sigma_{\sigma}.$$

Let π_N be the invariant law from Theorem 7.4. Then for every bounded measurable φ ,

$$\begin{aligned} \int \varphi d(\sigma_{\#} \pi_N) &= \int \varphi \circ \Sigma_{\sigma} d\pi_N \\ &= \int P_t^N(\varphi \circ \Sigma_{\sigma}) d\pi_N \\ &= \int (P_t^N \varphi) \circ \Sigma_{\sigma} d\pi_N \\ &= \int P_t^N \varphi d(\sigma_{\#} \pi_N). \end{aligned}$$

Thus $\sigma_{\#} \pi_N$ is another invariant probability measure for the same semigroup. By the uniqueness of the invariant law π_N , $\sigma_{\#} \pi_N = \pi_N$. $\square$

Theorem 7.6 (Finite de Finetti representation). *For each $N \geq 2$ there exists a probability measure Q_N on $\mathcal{P}(\mathbb{R}^{2d})$ such that, for every $1 \leq k \leq N$,*

$$d_{\text{TV}}\left(\pi_{N,k}, \int_{\mathcal{P}(\mathbb{R}^{2d})} \mu^{\otimes k} Q_N(d\mu)\right) \leq \frac{k(k-1)}{2N},$$

where $\pi_{N,k}$ is the law of the first k particles under π_N . One may take Q_N to be the law of the empirical measure

$$L_N := \frac{1}{N} \sum_{i=1}^N \delta_{(X_i, V_i)} \quad \text{under } \pi_N.$$

Proof. The state space $\mathbb{R}^{2d}$ is a standard Borel space, and Theorem 7.5 shows that π_N is exchangeable on $(\mathbb{R}^{2d})^N$. The finite de Finetti theorem of Diaconis and Freedman [?] therefore applies directly. It gives a probability measure Q_N on $\mathcal{P}(\mathbb{R}^{2d})$ such that

$$d_{\text{TV}}\left(\pi_{N,k}, \int \mu^{\otimes k} Q_N(d\mu)\right) \leq \frac{k(k-1)}{2N} \quad \forall 1 \leq k \leq N.$$

For the canonical choice, define the empirical-measure map

$$\mathfrak{L}_N : (\mathbb{R}^{2d})^N \rightarrow \mathcal{P}(\mathbb{R}^{2d}), \quad \mathfrak{L}_N(z_1, \dots, z_N) := \frac{1}{N} \sum_{i=1}^N \delta_{z_i},$$

and set $Q_N := (\mathfrak{L}_N)_\# \pi_N$. This is exactly the law of L_N under π_N , and the same Diaconis-Freedman theorem gives the stated bound for this canonical mixing measure. $\square$

Corollary 7.7 (Exact barycentric representation of the first marginal). *If $\pi_{N,1}$ denotes the first marginal of π_N , then*

$$\pi_{N,1} = \int_{\mathcal{P}(\mathbb{R}^{2d})} \mu Q_N(d\mu).$$

Proof. Apply Theorem 7.6 with $k = 1$. Since

$$\frac{1(1-1)}{2N} = 0,$$

the total-variation distance is zero:

$$d_{\text{TV}}\left(\pi_{N,1}, \int \mu Q_N(d\mu)\right) = 0.$$

Two probability measures at total-variation distance zero are equal, so the first marginal is exactly the barycenter of Q_N . $\square$

Lemma 7.8 (Conditioning on the empirical measure). *Let $G : \mathbb{R}^{2d} \times \mathcal{P}(\mathbb{R}^{2d}) \rightarrow \mathbb{R}$ be bounded and measurable. Under the exchangeable law π_N ,*

$$\mathbb{E}_{\pi_N}[G((X_1, V_1), L_N)] = \mathbb{E}_{\pi_N}\left[\int_{\mathbb{R}^{2d}} G(z, L_N) L_N(dz)\right].$$

Proof. By exchangeability, each index i has the same law as index 1, so

$$\mathbb{E}_{\pi_N}[G((X_1, V_1), L_N)] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\pi_N}[G((X_i, V_i), L_N)].$$

Inside the expectation, the empirical average is exactly the integral against the empirical measure:

$$\frac{1}{N} \sum_{i=1}^N G((X_i, V_i), L_N) = \int_{\mathbb{R}^{2d}} G(z, L_N) L_N(dz).$$

Taking expectations of both sides gives the claimed identity. $\square$

7.3 Tightness, identification, and the stationary thermodynamic limit

Theorem 7.9 (Tightness of the stationary marginals). *The sequence of first marginals $\{\pi_{N,1}\}_{N \geq 2}$ is tight in $\mathcal{P}(\mathbb{R}^{2d})$.*

Proof. By Theorem 7.4, the invariant law π_N satisfies (30). Using the equivalence (28), there is a constant $C_2 < \infty$, independent of N , such that

$$\int \frac{1}{N} \sum_{i=1}^N (\|x_i\|^2 + \|v_i\|^2) \pi_N(dX dV) \leq C_2.$$

Using exchangeability from Theorem 7.5, this becomes

$$\int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) \pi_{N,1}(dx dv) \leq C_2 \quad \forall N \geq 2.$$

Now fix $\varepsilon > 0$ and choose $R_\varepsilon := \sqrt{C_2/\varepsilon}$. Markov's inequality gives

$$\pi_{N,1}(\{(x, v) : \|x\|^2 + \|v\|^2 > R_\varepsilon^2\}) \leq \frac{C_2}{R_\varepsilon^2} = \varepsilon.$$

Hence the compact ball $K_\varepsilon := \{(x, v) : \|x\|^2 + \|v\|^2 \leq R_\varepsilon^2\}$ satisfies

$$\pi_{N,1}(K_\varepsilon) \geq 1 - \varepsilon \quad \forall N \geq 2.$$

Prokhorov's theorem therefore yields tightness. $\square$

7.4 Linear stationary problem and nonlinear alignment fixed point

Proposition 7.10 (Linear stationary law for a prescribed alignment field). *Let $a : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be bounded and globally Lipschitz, and define*

$$\mathcal{L}^a \phi(x, v) := v \cdot \nabla_x \phi + (-\nabla U(x) - (\gamma + \nu_{\text{visc}})v + \nu_{\text{visc}} a(x)) \cdot \nabla_v \phi + \frac{\sigma^2}{2} \Delta_v \phi.$$

Then the diffusion generated by $\mathcal{L}^a$ admits a unique invariant law $\pi^a \in \mathcal{P}(\mathbb{R}^{2d})$. Moreover:

- (i) π^a has finite second moments, with a bound depending only on $\|a\|_\infty$ and the structural constants;
- (ii) π^a admits a smooth strictly positive density ρ^a ;

(iii) if a ranges over a bounded set in $W^{1,\infty}$, then the second-moment bound is uniform on that set.

Proof. The drift of $\mathcal{L}^a$ is the confining kinetic Langevin drift $(v, -\nabla U(x) - \gamma v)$ plus the bounded perturbation $\nu_{\text{visc}}(a(x) - v)$ in velocity. The same quadratic cross-term Lyapunov function used in Theorem 7.4 yields

$$\mathcal{L}^a W(x, v) \leq -c_0 W(x, v) + C_0(1 + \|a\|_\infty^2)$$

for a quadratic $W \sim 1 + \|x\|^2 + \|v\|^2$. The diffusion matrix is constant in velocity and the coefficients are smooth, so the bracket family generated by $\{\partial_{v_1}, \dots, \partial_{v_d}\}$ and the drift field spans the full tangent space:

$$[v \cdot \nabla_x + (-\nabla U(x) - (\gamma + \nu_{\text{visc}})v + \nu_{\text{visc}}a(x)) \cdot \nabla_v, \partial_{v_k}] = \partial_{x_k} - (\gamma + \nu_{\text{visc}})\partial_{v_k}.$$

Hence Hörmander's condition holds globally. The transition kernel therefore has a smooth strictly positive density for every $t > 0$, and every compact set is small by the same compact-ball argument used in the finite-particle minorization theorem. Harris theory then gives existence and uniqueness of the invariant probability law π^a together with the stated quadratic moment bound. Finally, invariance and the strictly positive smooth transition density imply that π^a itself admits a smooth strictly positive density ρ^a . $\square$

Definition 7.11 (Stationary alignment map). For a bounded Lipschitz field a , let ρ^a be the density of the invariant law π^a from Proposition 7.10. Define

$$(\mathcal{T}a)(x) := \frac{\iint_{\mathbb{R}^{2d}} K(x, y) q \rho^a(y, q) \, dy \, dq}{\iint_{\mathbb{R}^{2d}} K(x, y) \rho^a(y, q) \, dy \, dq}.$$

Lemma 7.12 (Invariant ball for the alignment map). *There exists $R_* > 0$, depending only on the structural constants, such that if*

$$\mathcal{A}_{R_*} := \{a \in W^{1,\infty}(\mathbb{R}^d; \mathbb{R}^d) : \|a\|_\infty + \|\nabla a\|_\infty \leq R_*\},$$

then $\mathcal{T}(\mathcal{A}_{R_}) \subseteq \mathcal{A}_{R_*}$.*

Proof. By the kernel floor, the denominator in Definition 7.11 is bounded below by k_* . The numerator is controlled by the first velocity moment of π^a , which is uniformly bounded on bounded $W^{1,\infty}$ balls by Proposition 7.10. Differentiating under the integral sign and using the boundedness of ∇K yields the same control for $\nabla(\mathcal{T}a)$. Choosing R_* larger than the resulting constant closes the invariant-ball estimate. $\square$

Lemma 7.13 (Small-viscosity contraction of the alignment map). *There exists $C_{\text{mf}} < \infty$, depending only on the structural constants, such that for all $a, b \in \mathcal{A}_{R_*}$,*

$$\|\mathcal{T}a - \mathcal{T}b\|_{L^\infty} \leq C_{\text{mf}} \frac{\nu_{\text{visc}}}{\gamma} \|a - b\|_{L^\infty}.$$

Consequently, if

$$0 \leq \nu_{\text{visc}} < \frac{\gamma}{C_{\text{mf}}},$$

then $\mathcal{T}$ is a strict contraction on $\mathcal{A}_{R_}$.*

Proof. Let (X_t^a, V_t^a) and (X_t^b, V_t^b) solve the two frozen-field SDEs driven by the same Brownian motion:

$$\begin{aligned} dX_t^\sharp &= V_t^\sharp dt, \\ dV_t^\sharp &= \left(-\nabla U(X_t^\sharp) - (\gamma + \nu_{\text{visc}})V_t^\sharp + \nu_{\text{visc}} \sharp(X_t^\sharp) \right) dt + \sigma dW_t, \quad \sharp \in \{a, b\}. \end{aligned}$$

Choose the initial law to be an arbitrary coupling of π^a and π^b . Since each marginal is started from its own invariant law, both marginals remain stationary for every $t \geq 0$. Set

$$\Delta X_t := X_t^a - X_t^b, \quad \Delta V_t := V_t^a - V_t^b.$$

Then

$$\begin{aligned} d\Delta X_t &= \Delta V_t dt, \\ d\Delta V_t &= \left(-(\nabla U(X_t^a) - \nabla U(X_t^b)) - (\gamma + \nu_{\text{visc}})\Delta V_t \right. \\ &\quad \left. + \nu_{\text{visc}}(a(X_t^a) - a(X_t^b)) + \nu_{\text{visc}}(a(X_t^b) - b(X_t^b)) \right) dt. \end{aligned}$$

Set

$$\beta := \frac{\gamma + \nu_{\text{visc}}}{2}, \quad \alpha := (\gamma + \nu_{\text{visc}})\beta = \frac{(\gamma + \nu_{\text{visc}})^2}{2},$$

and define

$$\mathcal{Q}(\xi, \eta) := \alpha \|\xi\|^2 + 2\beta \langle \xi, \eta \rangle + \|\eta\|^2.$$

Since $\alpha > \beta^2$, this quadratic form is equivalent to $\|\xi\|^2 + \|\eta\|^2$. For the unforced part of the difference dynamics, Itô's formula gives

$$\begin{aligned} \frac{d}{dt} \mathcal{Q}(\Delta X_t, \Delta V_t) &= 2(\alpha - (\gamma + \nu_{\text{visc}})\beta) \langle \Delta X_t, \Delta V_t \rangle + 2(\beta - (\gamma + \nu_{\text{visc}})) \|\Delta V_t\|^2 \\ &\quad - 2\beta \langle \Delta X_t, \nabla U(X_t^a) - \nabla U(X_t^b) \rangle - 2 \langle \Delta V_t, \nabla U(X_t^a) - \nabla U(X_t^b) \rangle \\ &\quad + 2\nu_{\text{visc}}\beta \langle \Delta X_t, a(X_t^a) - a(X_t^b) \rangle + 2\nu_{\text{visc}} \langle \Delta V_t, a(X_t^a) - a(X_t^b) \rangle \\ &\quad + 2\nu_{\text{visc}} \langle \Delta V_t + \beta \Delta X_t, a(X_t^b) - b(X_t^b) \rangle. \end{aligned}$$

By the choice of α , the cross term vanishes. Using the strong monotonicity and Lipschitz bounds on ∇U together with $\|\nabla a\|_\infty \leq R_*$, we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{Q}(\Delta X_t, \Delta V_t) &\leq -(\gamma + \nu_{\text{visc}}) \|\Delta V_t\|^2 - 2\beta(m_U - \nu_{\text{visc}}R_*) \|\Delta X_t\|^2 \\ &\quad + 2(L_U + \nu_{\text{visc}}R_*) \|\Delta X_t\| \|\Delta V_t\| \\ &\quad + 2\nu_{\text{visc}} \langle \Delta V_t + \beta \Delta X_t, a(X_t^b) - b(X_t^b) \rangle. \end{aligned}$$

Apply Young's inequality to the mixed term with $\varepsilon = (\gamma + \nu_{\text{visc}})/2$:

$$2(L_U + \nu_{\text{visc}}R_*) \|\Delta X_t\| \|\Delta V_t\| \leq \frac{\gamma + \nu_{\text{visc}}}{2} \|\Delta V_t\|^2 + \frac{2(L_U + \nu_{\text{visc}}R_*)^2}{\gamma + \nu_{\text{visc}}} \|\Delta X_t\|^2.$$

Hence

$$\frac{d}{dt} \mathcal{Q}(\Delta X_t, \Delta V_t) \leq -\frac{\gamma + \nu_{\text{visc}}}{2} \|\Delta V_t\|^2$$

$$\begin{aligned}
& - \left((\gamma + \nu_{\text{visc}})(m_U - \nu_{\text{visc}} R_*) - \frac{2(L_U + \nu_{\text{visc}} R_*)^2}{\gamma + \nu_{\text{visc}}} \right) \|\Delta X_t\|^2 \\
& + 2\nu_{\text{visc}} \langle \Delta V_t + \beta \Delta X_t, a(X_t^b) - b(X_t^b) \rangle.
\end{aligned}$$

In the small-viscosity / large-friction regime, the bracketed coefficient is strictly positive. By the equivalence of $\mathcal{Q}$ with $\|\Delta X_t\|^2 + \|\Delta V_t\|^2$, there exists $\lambda_{\text{cpl}} > 0$, depending only on (m_U, L_U, γ, R_*) , such that

$$\frac{d}{dt} \mathcal{Q}(\Delta X_t, \Delta V_t) \leq -\lambda_{\text{cpl}} \mathcal{Q}(\Delta X_t, \Delta V_t) + 2\nu_{\text{visc}} \langle \Delta V_t + \beta \Delta X_t, a(X_t^b) - b(X_t^b) \rangle. \quad (31)$$

Taking expectations in (31) and using Young's inequality gives

$$\frac{d}{dt} \mathbb{E} \mathcal{Q}(\Delta X_t, \Delta V_t) \leq -\frac{\lambda_{\text{cpl}}}{2} \mathbb{E} \mathcal{Q}(\Delta X_t, \Delta V_t) + C_0 \frac{\nu_{\text{visc}}^2}{\lambda_{\text{cpl}}} \|a - b\|_{L^\infty}^2.$$

Hence

$$\mathbb{E} \mathcal{Q}(\Delta X_t, \Delta V_t) \leq e^{-\lambda_{\text{cpl}} t/2} \mathbb{E} \mathcal{Q}(\Delta X_0, \Delta V_0) + C_1 \frac{\nu_{\text{visc}}^2}{\lambda_{\text{cpl}}} \|a - b\|_{L^\infty}^2.$$

Because both marginals are stationary, the law of $(X_t^a, V_t^a, X_t^b, V_t^b)$ is a coupling of π^a and π^b for every t . Letting $t \rightarrow \infty$ and using the equivalence of $\mathcal{Q}$ with $\|\Delta X_t\|^2 + \|\Delta V_t\|^2$, we obtain

$$W_2(\pi^a, \pi^b) \leq C_2 \frac{\nu_{\text{visc}}}{\lambda_{\text{cpl}}} \|a - b\|_{L^\infty}. \quad (32)$$

The same quadratic-form construction gives $\lambda_{\text{cpl}} \gtrsim \gamma$ uniformly in the small-viscosity regime, so

$$W_2(\pi^a, \pi^b) \leq C_3 \frac{\nu_{\text{visc}}}{\gamma} \|a - b\|_{L^\infty}.$$

Fix $x \in \mathbb{R}^d$. Write

$$N_a(x) := \iint K(x, y) q \pi^a(dy dq), \quad D_a(x) := \iint K(x, y) \pi^a(dy dq),$$

so that $(\mathcal{T}a)(x) = N_a(x)/D_a(x)$, and similarly for b . Since $K \geq k_*$, we have $D_a(x), D_b(x) \geq k_*$. Proposition 7.10 gives a uniform second-moment bound on π^a, π^b over $\mathcal{A}_{R_*}$. Therefore, for any coupling Π of π^a and π^b ,

$$\begin{aligned}
\|N_a(x) - N_b(x)\| & \leq \iint \|K(x, y)q - K(x, y')q'\| \Pi(dy dq dy' dq') \\
& \leq k^* \iint \|q - q'\| d\Pi + L_K \left(\iint \|q\|^2 d\Pi \right)^{1/2} \left(\iint \|y - y'\|^2 d\Pi \right)^{1/2} \\
& \leq C_4 W_2(\pi^a, \pi^b).
\end{aligned}$$

Likewise,

$$|D_a(x) - D_b(x)| \leq L_K W_2(\pi^a, \pi^b).$$

Using

$$\frac{N_a}{D_a} - \frac{N_b}{D_b} = \frac{N_a - N_b}{D_a} + N_b \frac{D_b - D_a}{D_a D_b},$$

the denominator lower bound k_* , and the uniform first-moment bound on π^b , we conclude that

$$\|(\mathcal{T}a)(x) - (\mathcal{T}b)(x)\| \leq C_5 W_2(\pi^a, \pi^b) \leq C_{\text{mf}} \frac{\nu_{\text{visc}}}{\gamma} \|a - b\|_{L^\infty}.$$

Taking the supremum over x yields the claim. $\square$

Proposition 7.14 (Unique stationary mean-field law). *There exists a unique stationary probability law $\pi_\infty \in \mathcal{P}(\mathbb{R}^{2d})$ for the mean-field equation (5). Equivalently, π_∞ is the unique probability law such that*

$$\int_{\mathbb{R}^{2d}} \mathcal{L}_{\pi_\infty} \phi(z) \pi_\infty(dz) = 0 \quad \forall \phi \in C_c^\infty(\mathbb{R}^{2d}), \quad (33)$$

where

$$\mathcal{L}_\mu \phi(x, v) := v \cdot \nabla_x \phi + (-\nabla U(x) - \gamma v + F[\mu](x, v)) \cdot \nabla_v \phi + \frac{\sigma^2}{2} \Delta_v \phi.$$

Proof. Fix the invariant ball $\mathcal{A}_{R_*}$ from Lemma 7.12. If

$$0 \leq \nu_{\text{visc}} < \frac{\gamma}{C_{\text{mf}}},$$

then Lemma 7.13 shows that $\mathcal{T}$ is a strict contraction on the complete metric space $\mathcal{A}_{R_*}$. Banach's fixed-point theorem therefore yields a unique field $a_\infty \in \mathcal{A}_{R_*}$ such that

$$\mathcal{T}a_\infty = a_\infty.$$

Let π^{a_∞} be the invariant law of the linear problem from Proposition 7.10. By the definition of $\mathcal{T}$, the associated density ρ^{a_∞} satisfies

$$\nu_{\text{visc}}(a_\infty(x) - v) = F[\pi^{a_\infty}](x, v),$$

so π^{a_∞} satisfies exactly the nonlinear stationary weak equation (33). This gives existence of a stationary mean-field law $\pi_\infty := \pi^{a_\infty}$. Conversely, any stationary solution $\bar{\pi}$ induces through its alignment profile a fixed point of $\mathcal{T}$, hence by uniqueness of the fixed point must coincide with π_∞ . $\square$

Theorem 7.15 (Stationary propagation of chaos). *The sequence of first marginals $\pi_{N,1}$ converges weakly to the unique stationary mean-field law π_∞ from Proposition 7.14. Equivalently,*

$$\pi_{N,1} \Rightarrow \pi_\infty \quad \text{as } N \rightarrow \infty.$$

Proof. We prove tightness, identify every subsequential limit, and use uniqueness of the stationary mean-field law to conclude convergence.

Step 1: tightness. By Theorem 7.9, the family $\{\pi_{N,1}\}_{N \geq 2}$ is tight. Hence every subsequence admits a weakly convergent sub-subsequence.

Step 2: stationarity identity for the finite- N system. Fix $\phi \in C_c^\infty(\mathbb{R}^{2d})$ and set

$$\Phi(X_1, V_1, \dots, X_N, V_N) := \phi(X_1, V_1).$$

Since π_N is invariant,

$$\int \mathcal{L}_N \Phi d\pi_N = 0.$$

Writing

$$\mathcal{L}_\mu \phi(x, v) := v \cdot \nabla_x \phi + (-\nabla U(x) - \gamma v + F[\mu](x, v)) \cdot \nabla_v \phi + \frac{\sigma^2}{2} \Delta_v \phi,$$

this gives

$$\int \mathcal{L}_0 \phi(z_1) \pi_N(dZ) + \int F_1^N(Z) \cdot \nabla_v \phi(z_1) \pi_N(dZ) = 0,$$

where $\mathcal{L}_0$ is the backbone generator without the mean-field term. By Lemma 6.1, evaluated under the stationary law π_N ,

$$\left| \int (F_1^N - F[L_N](z_1)) \cdot \nabla_v \phi(z_1) d\pi_N \right| \leq \|\nabla_v \phi\|_\infty \left(\int \|F_1^N - F[L_N](z_1)\|^2 d\pi_N \right)^{1/2} \leq \frac{C_\phi}{\sqrt{N}}.$$

$$\Psi_\phi(\mu) := \int_{\mathbb{R}^{2d}} \mathcal{L}_\mu \phi(z) \mu(dz).$$

By Corollary 7.7 and Lemma 7.8,

$$|\mathbb{E}_{\pi_N}[\Psi_\phi(L_N)]| \leq \frac{C_\phi}{\sqrt{N}}. \quad (34)$$

Step 4: identification of limit points. Let $\{N_k\}$ be a subsequence such that $\pi_{N_k,1} \Rightarrow \bar{\pi}$. By the uniform second-moment bound in Theorem 7.9 and the canonical choice $Q_N = \text{Law}(L_N)$ from Theorem 7.6, the family $\{Q_N\}_{N \geq 2}$ is tight in $\mathcal{P}(\mathcal{P}(\mathbb{R}^{2d}))$. Passing to a further subsequence if necessary, we may assume $Q_{N_k} \Rightarrow Q_\infty$. Indeed,

$$\int_{\mathcal{P}(\mathbb{R}^{2d})} \left[\int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) \mu(dx dv) \right] Q_N(d\mu) = \int_{\mathbb{R}^{2d}} (\|x\|^2 + \|v\|^2) \pi_{N,1}(dx dv) \leq C_2,$$

so the same Markov-Prokhorov argument applies on the second level. For compactly supported ϕ , the map Ψ_ϕ is continuous on $\mathcal{P}(\mathbb{R}^{2d})$ under weak convergence together with uniform second moments, because K and $\nabla_v \phi$ are bounded and the denominator in $F[\mu]$ is uniformly bounded below by k_* . Passing to the limit in the previous display therefore gives

$$\int_{\mathcal{P}(\mathbb{R}^{2d})} \Psi_\phi(\mu) Q_\infty(d\mu) = 0 \quad \forall \phi \in C_c^\infty(\mathbb{R}^{2d}).$$

This is the stationary weak mean-field identity for the limit mixture. The vanishing defect estimate (34), together with the continuity just noted, puts us in the hypotheses of the standard nonlinear martingale-problem identification for exchangeable particle systems (equivalently, the de Finetti/Sznitman limit-point argument; see [?]). Hence

$$Q_\infty \left(\left\{ \mu : \int_{\mathbb{R}^{2d}} \mathcal{L}_\mu \phi(z) \mu(dz) = 0 \quad \forall \phi \in C_c^\infty(\mathbb{R}^{2d}) \right\} \right) = 1.$$

By Proposition 7.14, this set consists of the single law π_∞ , so $Q_\infty = \delta_{\pi_\infty}$.

Step 5: uniqueness and collapse of the mixing measure. Since the barycenter map is continuous on the uniform second-moment class,

$$\bar{\pi} = \int \mu Q_\infty(d\mu) = \int \mu \delta_{\pi_\infty}(d\mu) = \pi_\infty.$$

Thus every convergent subsequence of $\{\pi_{N,1}\}_{N \geq 2}$ has the same limit, so the whole sequence converges:

$$\pi_{N,1} \Rightarrow \pi_\infty.$$

□

Corollary 7.16 (Thermodynamic limit for bounded observables). *For every bounded continuous observable $\phi : \mathbb{R}^{2d} \rightarrow \mathbb{R}$,*

$$\lim_{N \rightarrow \infty} \mathbb{E}_{\pi_N} \left[\frac{1}{N} \sum_{i=1}^N \phi(X_i, V_i) \right] = \int_{\mathbb{R}^{2d}} \phi d\pi_\infty.$$

Proof. Exchangeability gives

$$\mathbb{E}_{\pi_N} \left[\frac{1}{N} \sum_{i=1}^N \phi(X_i, V_i) \right] = \int_{\mathbb{R}^{2d}} \phi d\pi_{N,1}.$$

Now apply Theorem 7.15 and the definition of weak convergence. □

Definition 7.17 (Backbone hypocoercive coefficients for the mean-field LSI). Set

$$M := L_U, \quad \varepsilon := \min \left\{ 1, \frac{\gamma}{16(M + \gamma)^2} \right\},$$

and define

$$a := \frac{\gamma\varepsilon}{4}, \quad b := \varepsilon, \quad c := \varepsilon^2.$$

These are exactly the coefficients from Theorem 4.5, recorded here so that the mean-field functional-inequality statement can be read independently of the preceding backbone section.

Proposition 7.18 (Applicability of Huang–Kopfer–Ren to the frozen mean-field generator). *Let $a_\infty \in \mathcal{A}_{R_*}$ be the unique fixed point from Proposition 7.14, and let*

$$\mathcal{L}^{a_\infty} \phi := v \cdot \nabla_x \phi + (-\nabla U(x) - (\gamma + \nu_{\text{visc}})v + \nu_{\text{visc}} a_\infty(x)) \cdot \nabla_v \phi + \frac{\sigma^2}{2} \Delta_v \phi.$$

Then, in the notation of Huang–Kopfer–Ren [?, Theorem 3.5], the coefficients of $\mathcal{L}^{a_\infty}$ satisfy:

- (i) *the measure-interaction constant is $K_I = 0$;*
- (ii) *the diffusion bounds [?, (57), (63)] hold with $\delta_1 = \delta_2 = \sigma^2$;*
- (iii) *the dissipativity condition [?, (C) with $R = 0$] holds.*

Consequently, Huang–Kopfer–Ren’s frozen-measure logarithmic Sobolev theorem applies to the invariant law $\pi_\infty = \pi^{a_\infty}$.

Proof. Item (i) is immediate: the field a_∞ is frozen, so the generator $\mathcal{L}^{a_\infty}$ is an ordinary kinetic diffusion with no remaining dependence on the current law. Hence the interaction parameter K_I in [?] is exactly zero.

For item (ii), the diffusion acts only in the velocity variables and is constant:

$$\Sigma := \begin{pmatrix} 0 \\ \sigma I_d \end{pmatrix}, \quad \Sigma \Sigma^\top = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 I_d \end{pmatrix}.$$

Therefore the non-degenerate block in the diffusive coordinates is exactly $\sigma^2 I_d$, so the upper and lower ellipticity bounds required in [?, (57), (63)] hold with the same constant σ^2 .

For item (iii), consider two solutions of the frozen-field SDE driven by the same Brownian motion and write their difference as $(\Delta X_t, \Delta V_t)$. The proof of Lemma 7.13, specialized to the case $a = b = a_\infty$, shows that the forcing term disappears and the quadratic form

$$\mathcal{Q}(\xi, \eta) = \alpha \|\xi\|^2 + 2\beta \langle \xi, \eta \rangle + \|\eta\|^2$$

satisfies

$$\frac{d}{dt} \mathcal{Q}(\Delta X_t, \Delta V_t) \leq -\lambda_{\text{cpl}} \mathcal{Q}(\Delta X_t, \Delta V_t)$$

for some $\lambda_{\text{cpl}} > 0$ depending only on the structural constants. Equivalently, the drift of $\mathcal{L}^{a_\infty}$ is uniformly dissipative with respect to the positive definite metric induced by $\mathcal{Q}$. This is exactly the content of [?, (C) with $R = 0$] in the frozen, interaction-free case.

Thus all hypotheses of [?, Theorem 3.5] are satisfied for the invariant law of $\mathcal{L}^{a_\infty}$, namely π_∞ . $\square$

Theorem 7.19 (Perturbative stationary mean-field LSI). *Let*

$$\Gamma_{\text{hypo}}^\infty(g, g) := a \|\nabla_v g\|^2 + 2b \nabla_x g \cdot \nabla_v g + c \|\nabla_x g\|^2, \quad ac > b^2,$$

with the coefficients a, b, c from Definition 7.17. Assume in addition that

$$0 \leq \nu_{\text{visc}} \leq \nu_*,$$

with ν_ chosen small enough that Proposition 7.14 applies. Then the stationary mean-field law π_∞ satisfies*

$$\text{Ent}_{\pi_\infty}(g^2) \leq C_{\text{LSI}} \int_{\mathbb{R}^{2d}} \Gamma_{\text{hypo}}^\infty(g, g) \, d\pi_\infty$$

for every smooth compactly supported g , for some constant $C_{\text{LSI}} < \infty$ depending only on the structural constants.

Proof. At this stage the stationary mean-field law π_∞ has already been constructed and shown to be unique in the perturbative small-viscosity regime by Proposition 7.14. The remaining task is therefore to verify that the frozen generator associated with π_∞ satisfies the hypotheses of the imported frozen-measure logarithmic Sobolev criterion. Proposition 7.18 provides exactly that verification in the present notation.

Proposition 7.18 shows that the frozen-field kinetic generator associated with π_∞ satisfies the exact hypotheses of Huang–Kopfer–Ren’s frozen-measure logarithmic Sobolev theorem [?, Theorem 3.5]. Hence there exists a constant $C_E < \infty$, depending only on the structural bounds of the frozen field ball $\mathcal{A}_{R_*}$, such that

$$\text{Ent}_{\pi_\infty}(g^2) \leq C_E \int_{\mathbb{R}^{2d}} (\|\nabla_x g\|^2 + \|\nabla_v g\|^2) \, d\pi_\infty$$

for every smooth compactly supported g .

Now let

$$M_\Gamma := \begin{pmatrix} c & b \\ b & a \end{pmatrix}.$$

Since $ac > b^2$, the matrix M_Γ is positive definite. Its smallest eigenvalue is

$$\lambda_\Gamma := \frac{a + c - \sqrt{(a - c)^2 + 4b^2}}{2} > 0.$$

Consequently,

$$\Gamma_{\text{hypo}}^\infty(g, g) \geq \lambda_\Gamma (\|\nabla_x g\|^2 + \|\nabla_v g\|^2).$$

Combining the two inequalities gives

$$\text{Ent}_{\pi_\infty}(g^2) \leq \frac{C_E}{\lambda_\Gamma} \int_{\mathbb{R}^{2d}} \Gamma_{\text{hypo}}^\infty(g, g) \, d\pi_\infty.$$

Renaming C_E/λ_Γ as C_{LSI} proves the claimed perturbative stationary mean-field LSI. $\square$

8 Conclusion

The main line of the paper establishes a comprehensive finite-time and stationary mean-field analysis for an underdamped diffusion with row-normalized alignment. Unconditionally under the standing assumptions, we prove global well-posedness for the particle system and the McKean–Vlasov equation, propagation of quadratic moments, a quantitative finite-time mean-field limit, and the existence of finite-particle invariant laws together with stationary thermodynamic compactness.

The perturbative small-viscosity regime enters only in the stationary uniqueness and functional-inequality stage. In that regime, the alignment-map contraction yields uniqueness of the stationary McKean–Vlasov law, and the frozen-generator criterion yields a stationary mean-field logarithmic Sobolev inequality.

The main analytic novelty is the treatment of the row-normalized denominator. The strict kernel floor prevents singular normalization, the self-interaction defect is of order N^{-1} , and this makes the normalized alignment compatible with both quantitative finite-time propagation of chaos and the stationary thermodynamic limit.

The appendix records translation-covariance and recentering statements for translated confining families. Those results are independent of the main finite-time and stationary mean-field theory and may be read separately.

A Translated Confinement and the Wall-to-Infinity Limit

This appendix records auxiliary covariance and recentering statements that are independent of the main finite-time and stationary mean-field results.

Definition A.1 (Translated confining family). Fix a reference potential $U_* : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfying Assumption 2.1. For each translation vector $a \in \mathbb{R}^d$, define

$$U_a(x) := U_*(x - a).$$

Let $P_t^{N,a}$, $Q_\tau^{N,a}$, π_N^a , and π_∞^a denote, respectively, the continuous finite-particle semigroup, the squashed observation chain, the invariant law, and the stationary mean-field law for the kinetic core with confining potential U_a .

Define the translation maps

$$t_a(x, v) := (x + a, v) \in \mathbb{R}^{2d},$$

$$T_a^N(x_1, v_1, \dots, x_N, v_N) := (x_1 + a, v_1, \dots, x_N + a, v_N) \in (\mathbb{R}^{2d})^N.$$

Theorem A.2 (Translation covariance of the confining family). *For every $a \in \mathbb{R}^d$, the translated family from Definition A.1 satisfies the following exact covariance relations.*

(i) If $(X_i^0(t), V_i^0(t))_{i=1}^N$ solves (1)–(2) with potential U_* , then

$$X_i^a(t) := X_i^0(t) + a, \quad V_i^a(t) := V_i^0(t)$$

solves the same system with potential U_a , driven by the same Brownian motions.

(ii) The continuous semigroups and squashed observation chains satisfy

$$P_t^{N,a}(F \circ T_a^N) = (P_t^{N,0}F) \circ T_a^N, \quad Q_\tau^{N,a}(F \circ T_a^N) = (Q_\tau^{N,0}F) \circ T_a^N$$

for every bounded measurable F .

(iii) The invariant laws satisfy

$$T_{a\#}^N \pi_N^0 = \pi_N^a.$$

(iv) If f_t^0 solves the mean-field equation (5) with potential U_* , then

$$f_t^a := t_{a\#} f_t^0$$

solves the translated mean-field equation with potential U_a . In particular,

$$t_{a\#} \pi_\infty^0 = \pi_\infty^a.$$

(v) The constants appearing in the stationary mean-field logarithmic Sobolev inequality from Theorem 7.19, as well as the stationary moment and compactness bounds used in the thermodynamic limit, are the same for every a .

Proof. The kernel is translation invariant:

$$K(x_i + a, x_j + a) = K(x_i, x_j).$$

Hence

$$d_i(X + a\mathbf{1}) = d_i(X), \quad \omega_{ij}(X + a\mathbf{1}) = \omega_{ij}(X),$$

and therefore the viscous force is translation covariant:

$$F_i(X + a\mathbf{1}, V) = F_i(X, V).$$

The confining drift transforms as

$$\nabla U_a(x + a) = \nabla U_*(x).$$

Substituting these identities into (1)–(2) shows that if (X_i^0, V_i^0) solves the reference system, then $(X_i^0 + a, V_i^0)$ solves the translated system. This proves (i).

For (ii), let $Z_t^{N,0}$ be the reference process and $Z_t^{N,a} := T_a^N Z_t^{N,0}$. By (i), $Z_t^{N,a}$ has the law of the translated process started from $T_a^N z$ whenever $Z_t^{N,0}$ starts from z . Therefore

$$P_t^{N,a}(F \circ T_a^N)(z) = \mathbb{E}_z[F(T_a^N Z_t^{N,a})] = \mathbb{E}_{T_a^N z}[F(Z_t^{N,0})] = (P_t^{N,0}F)(T_a^N z).$$

The observation map $S(x, v) = (x, \psi(v))$ commutes with translations in the position variables:

$$S \circ T_a^N = T_a^N \circ S.$$

Hence the same covariance relation holds for the observed chain $Q_\tau^{N,a}$.

For (iii), let $\mu := T_{a\#}^N \pi_N^0$. Then for every bounded measurable F ,

$$\int P_t^{N,a} F d\mu = \int P_t^{N,a} F(T_a^N z) \pi_N^0(dz) = \int P_t^{N,0} (F \circ T_a^N)(z) \pi_N^0(dz) = \int F(T_a^N z) \pi_N^0(dz) = \int F d\mu,$$

where the third equality uses invariance of π_N^0 under $P_t^{N,0}$. Thus μ is invariant for the translated dynamics. By the uniqueness part of Theorem 7.4, $\mu = \pi_N^a$.

For (iv), let f_t^0 be a weak solution of (5) with potential U_* , and define $f_t^a = t_{a\#} f_t^0$. Because K is translation invariant, the normalized kernel mass transforms by

$$m_{f_t^a}(x+a) = m_{f_t^0}(x),$$

and therefore

$$F[f_t^a](x+a, v) = F[f_t^0](x, v).$$

Writing the translated mean-field SDE with potential U_a ,

$$d\bar{X}_t^a = \bar{V}_t^a dt, \quad d\bar{V}_t^a = (-\nabla U_a(\bar{X}_t^a) - \gamma \bar{V}_t^a + F[f_t^a](\bar{X}_t^a, \bar{V}_t^a)) dt + \sigma dW_t,$$

and setting $(\bar{X}_t^a, \bar{V}_t^a) := (\bar{X}_t^0 + a, \bar{V}_t^0)$ gives exactly the reference SDE with potential U_* . Hence $f_t^a = t_{a\#} f_t^0$ is the translated mean-field solution. Applying this to the unique stationary law from Theorem 7.15 yields $t_{a\#} \pi_\infty^0 = \pi_\infty^a$.

For (v), the pointwise derivatives are unchanged by translation in the position variables. Consequently, for every smooth g ,

$$\Gamma_{\text{hypo}}^N(g \circ T_a^N, g \circ T_a^N) = \Gamma_{\text{hypo}}^N(g, g) \circ T_a^N,$$

and similarly for the mean-field carré du champ $\Gamma_{\text{hypo}}^\infty$. Using the pushforward identities from (iii) and (iv),

$$\text{Ent}_{\pi_N^a}(g^2) = \text{Ent}_{\pi_N^0}((g \circ T_a^N)^2), \quad \int \Gamma_{\text{hypo}}^N(g, g) d\pi_N^a = \int \Gamma_{\text{hypo}}^N(g \circ T_a^N, g \circ T_a^N) d\pi_N^0,$$

and likewise

$$\text{Ent}_{\pi_\infty^a}(g^2) = \text{Ent}_{\pi_\infty^0}((g \circ t_a)^2), \quad \int \Gamma_{\text{hypo}}^\infty(g, g) d\pi_\infty^a = \int \Gamma_{\text{hypo}}^\infty(g \circ t_a, g \circ t_a) d\pi_\infty^0.$$

Therefore the finite- N and mean-field logarithmic Sobolev inequalities for the translated family are identical to those of the reference family, with the same constants. The entropy contraction constants are the same because the semigroup covariance in (ii) conjugates the translated dynamics to the reference dynamics. $\square$

Corollary A.3 (Thermodynamic limit followed by wall-to-infinity recentering). *Let $(a_R)_{R \geq 1} \subset \mathbb{R}^d$ satisfy $\|a_R\| \rightarrow \infty$. For each R , consider the translated confining potential U_{a_R} . Then for every bounded continuous observable $\Phi : \mathbb{R}^{2d} \rightarrow \mathbb{R}$,*

$$\int_{\mathbb{R}^{2d}} \Phi(x - a_R, v) \pi_{N,1}^{a_R}(dx, dv) = \int_{\mathbb{R}^{2d}} \Phi(x, v) \pi_{N,1}^0(dx, dv)$$

for every N , and

$$\int_{\mathbb{R}^{2d}} \Phi(x - a_R, v) \pi_\infty^{a_R}(dx, dv) = \int_{\mathbb{R}^{2d}} \Phi(x, v) \pi_\infty^0(dx, dv)$$

for every R . Consequently the two limits commute:

$$\lim_{R \rightarrow \infty} \lim_{N \rightarrow \infty} \int \Phi(x - a_R, v) \pi_{N,1}^{a_R}(dx, dv) = \lim_{N \rightarrow \infty} \lim_{R \rightarrow \infty} \int \Phi(x - a_R, v) \pi_{N,1}^{a_R}(dx, dv) = \int \Phi(x, v) \pi_\infty^0(dx, dv).$$

Proof. The first identity is exactly the pushforward relation $(t_{-a_R})_{\#} \pi_{N,1}^{a_R} = \pi_{N,1}^0$, which follows from Theorem A.2(iii) by taking first marginals. The second identity is $(t_{-a_R})_{\#} \pi_{\infty}^{a_R} = \pi_{\infty}^0$, which is Theorem A.2(iv). The commutation of the $N \rightarrow \infty$ and $R \rightarrow \infty$ limits is then immediate, because after recentering the integral is independent of R , and the $N \rightarrow \infty$ limit is Theorem 7.15. $\square$